A Multi-Messenger Technosignature and Anomaly-Detection Campaign for Omega Centauri

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Abstract

Omega Centauri (NGC 5139), the most massive Galactic globular cluster and the probable stripped nucleus of an accreted dwarf galaxy, presents a unique conjunction of observational circumstances: the strongest current candidate for an intermediate-mass black hole (IMBH) in the Galaxy, anchored by seven stars moving above the local escape velocity (Häberle et al., 2024b); a formally unresolved factor-of-several tension between kinematic lower bounds ($\geq$ 8,200 $M_\odot$) and a pulsar-timing upper bound ($\lesssim$ 6,000 $M_\odot$; Bañares-Hernández et al. 2025); complete electromagnetic silence to the deepest radio and infrared limits ever placed on a globular cluster core (Mahida et al., 2026; Chen et al., 2026); and a southern declination optimal for the newest southern-hemisphere facilities. No dedicated technosignature search of ω Cen has ever been conducted at any wavelength. We present a coordinated, hypothesis-agnostic, multi-messenger campaign of eight instrument-matched programs (enumerated in Section 1) spanning infrared imaging, radio timing and SETI, astrometry, gravitational waves, neutrinos, gamma rays, the optical time domain, and archival channels, addressing conventional astrophysics (IMBH reality, mass, and spin; cluster dynamics) and technosignature hypotheses with the same data. For each program we state quantitative sensitivities, time requests, decision thresholds, and explicit falsification criteria, including negative results: direct astrometric acceleration detection of the fast stars is below 1σ at nominal parameters before ~2040, so the decision-grade astrometry routes through photocentric-wander and reference-frame measurements instead. Total cost is $\lesssim$ US\$10M over 2026–2035, most of it archival analysis and piggyback observing; the decisive mass and spin measurements arrive as by-products of planned LISA mission science. Every null result constrains conventional astrophysics, and no anomaly claim advances without confirmation from at least two independent messengers; under realistic outcomes the campaign adjudicates the astrophysical hypotheses, while the technosignature hypothesis is constrained only along specific low-probability branches.

Keywords: Omega Centauri; NGC 5139; intermediate-mass black holes; technosignatures; SETI; multi-messenger astronomy; millisecond pulsars; gravitational waves; neutrino astronomy

Contents

1 Introduction

1.1 The case for Omega Centauri now

Three independent developments between 2024 and 2026 have transformed Omega Centauri (ω Centauri, NGC 5139) from one interesting globular cluster among many into arguably the single best-posed multi-messenger target in the Galaxy outside the Galactic Centre.

First, the IMBH candidacy became concrete: Häberle et al. (2024b), using a proper-motion catalogue of 1.4 million stars built from over 500 HST epochs (Häberle et al., 2024a), identified seven stars within 3 arcsec of the cluster centre moving faster than the local escape velocity (five robust; two carried as a sensitivity row): stars that require a bound central mass of $\geq$ 8,200 $M_\odot$ from their velocities alone, and $\geq$ 21,100 $M_\odot$ at 99 per cent confidence, acceleration-model-informed, when acceleration limits are included.[1] Independent N-body modelling finds a best-fitting in-situ-grown IMBH of ~ 4.7–$5.1 \times 10^4\ M_\odot$ (González Prieto et al., 2025). Against this, a joint analysis of stellar kinematics and millisecond-pulsar (MSP) timing places a 3σ upper limit of $\sim 6 \times 10^3\ M_\odot$ on any central point mass, favouring instead an extended ≈ 2–$3 \times 10^5\ M_\odot$ component of stellar remnants (Bañares-Hernández et al., 2025). The constraints are formally inconsistent under their stated assumptions. A factor-of-several disagreement between two mature measurement techniques, centred on the nearest IMBH candidate in the sky, warrants a focused observational campaign to resolve it.

Second, the electromagnetic silence became quantitative: the most sensitive radio observation yet made of ω Cen ($\sim$170 h with ATCA, reaching 1.1 μJy beam^{-1} rms) detects nothing at any proposed cluster centre, bounding the accretion efficiency of a putative IMBH at $\lesssim 4 \times 10^{-3}$ of the Bondi rate (Mahida et al., 2026), extending two decades of radio non-detections of globular-cluster IMBHs (Strader et al., 2012; Tremou et al., 2018). JWST NIRCam and MIRI photometry of the central field likewise finds no source with the spectral energy distribution of an accreting IMBH (Chen et al., 2026). Whatever sits at the centre of ω Cen is dark to a depth that is a scientific result.

Third, the instrument landscape shifted decisively in the campaign's favour within an eighteen-month window centred on this writing: the Vera C. Rubin Observatory began science observations in late 2025 (the Pre-LSST period), with the ten-year Legacy Survey of Space and Time (LSST) formally begun on 2026 June 30 (Ivezić et al., 2019); the Nancy Grace Roman Space Telescope completed construction, with launch announced for 2026 August 30 (launch dates can move; we quote the announced date once and do not repeat the hedge) (WFIRST Astrometry Working Group et al., 2019); Gaia Data Release 4, with a 5.5-year astrometric baseline, is confirmed for 2026 December 2 (Gaia Collaboration et al., 2016); KM3NeT/ARCA reached 51 of 230 planned detection units (as of early 2026) with real-time alerts entering commissioning (Adrián-Martínez et al., 2016; The KM3NeT Collaboration, 2025); the first telescopes of CTAO-South are, as of early 2026, arriving at Paranal (Cherenkov Telescope Array Consortium et al., 2019); SKA-Mid is assembling toward science verification from 2027 and shared-risk operations from 2029 (Braun et al., 2019); ELT first light is scheduled for 2029 (Davies et al., 2021); and LISA, formally adopted by ESA in January 2024, is in hardware development for a 2035 launch (Colpi et al., 2024). Nearly every facility on this list is southern-hemisphere or all-sky; ω Cen, at declination $-47.5°$, sits in the sweet spot of all of them. A campaign organized now can ride this entire wave at marginal cost.

[1]The companion mass-tension analysis (Paper H) builds only on the firm 8,200 $M_\odot$ velocity-only bound from the five robust stars; it does not carry the 21,100 $M_\odot$ acceleration-informed figure, which depends on an acceleration model rather than kinematics alone.

1.2 The technosignature rationale

This paper is the observational companion to a speculative hypothesis paper (Swanson, 2026b, hereafter Paper A), which argues that rapidly spinning massive black holes in dense old stellar systems are thermodynamic attractors for hypothetical computation-optimizing civilizations, and that ω Cen is the most accessible system satisfying the resulting selection criteria. We neither assume nor argue for that hypothesis here. The campaign is designed to be *hypothesis-agnostic* in the specific sense that every observation in it is independently justified by conventional astrophysics: IMBH demographics (Greene et al., 2020), cluster dynamics, pulsar timing, accretion physics at the lowest Eddington ratios, and EMRI astrophysics (Amaro-Seoane, 2018). The technosignature dimension adds analysis channels (narrowband drift searches, burst-coincidence triggers, anomaly statistics) to data that would be taken anyway, in the cost-effective tradition of commensal SETI (Tarter, 2001; Wright et al., 2022; Lacki and DiKerby, 2025), consistent with the archival anomaly-detection strategy recommended by the Dyson Minds workshop for black-hole-hosted intelligence (Curtis et al., 2026), and aligned with recent calls for integrated, multi-channel technosignature-search strategies as the efficient way to constrain Fermi-paradox hypotheses (Crawford, 2026).

Three considerations nevertheless justify making the technosignature dimension explicit rather than incidental. First, globular clusters are almost entirely unexplored SETI territory: the first dedicated globular-cluster survey, a FAST pilot of five northern clusters, was published only this year and could not observe ω Cen from FAST's latitude (Huang et al., 2026b); Gaia-based dynamical metrics for prioritizing clusters as technosignature targets have followed (Huang et al., 2026a). Second, ω Cen is a privileged target on entirely generic arguments: 10^7 stars of age 12.08 Gyr (0.75 Gyr spread; systematics dominate the mean; Clontz et al. 2024) in a beam a few arcminutes across, a possible IMBH, and (as the probable nucleus of an accreted dwarf; Hilker and Richtler 2000; Ibata et al. 2019) an independent galactic chemical-evolution history, so that any ETI prior that weights stellar age and number density at all concentrates sharply here. Third, the high-energy technosignature channels (neutrino bursts from hypothetical black-hole-based computation; Dvali and Osmanov 2023) happen to be testable at ω Cen essentially for free, because the cluster's declination makes it an up-going source for the Mediterranean neutrino telescopes. Where Paper A's specific predictions sharpen a test, we say so explicitly and label the dependence; readers uninterested in that hypothesis may strike those sentences without loss to the campaign.

Epistemic status: The observational programs of Sections 3–10 stand on conventional astrophysics and are stated in the empirical register. The technosignature interpretation of any anomaly they return belongs to the speculative register, is conditional on the optimization premise of Papers A and B, and is adjudicated by the quantitative framework of Paper E rather than by this campaign.

1.3 Design principles

Five principles govern the campaign design, and we state them up front because they discipline everything that follows.

(i) Dual-use data. Every observing program must produce publishable conventional astrophysics on its own; the technosignature analysis is a parallel pipeline on the same photons (or neutrinos, or strain data), never a dedicated expenditure that a null result would waste.

(ii) Honest sensitivity accounting. Where a measurement cannot work at nominal parameters, we say so quantitatively. Section 5 demonstrates this standard: the direct acceleration test on the fast stars, superficially the most natural follow-up to Häberle et al. (2024b), is below 1σ before ~2040 for plausible masses, and we restructure the astrometric program accordingly.

(iii) Two-messenger adjudication. No anomaly claim survives on one channel. Every positive trigger (a neutrino multiplet, a narrowband line, an infrared transient) must specify in advance which independent channel confirms or kills it, with combined significance assessed by pre-registered methods (Section 11).

(iv) Pre-registered thresholds. Detection and falsification thresholds are stated numerically before the data arrive (Tables 6 and 5), following standard pre-registration practice.

(v) Parasitism on funded science. The decisive measurements (IMBH reality, mass, and spin) will be made by LISA as part of its core science program whether or not anyone organizes a campaign (Colpi et al., 2024; Babak et al., 2017). The campaign's job is to ensure that when those numbers arrive, the contextual data (timing, astrometry, electromagnetic limits, high-energy monitoring) already exist to interpret them.

1.4 Structure of this paper

Section 2 reviews the target and the present constraint landscape. Sections 3–10 present the eight programs, each with science case, technical implementation, sensitivity estimates, and decision criteria: (1) JWST infrared accretion and waste-heat limits; (2) deep radio imaging, narrowband SETI, and millisecond-pulsar timing with MeerKAT, transitioning to SKA-Mid; (3) astrometric monitoring with HST, Gaia DR4, the Nancy Grace Roman Space Telescope, and ELT/MICADO; (4) a LISA extreme-mass-ratio-inspiral forecast; (5) a KM3NeT/ARCA neutrino burst pipeline exploiting ω Cen's up-going geometry from the Mediterranean; (6) Fermi-LAT archival analysis and CTAO-South follow-up; (7) optical/infrared time-domain and archival anomaly searches with Rubin/LSST and oMEGACat; and (8) supplementary archival channels (LIGO–Virgo–KAGRA continuous waves; Pierre Auger ultra-high-energy cosmic rays). Section 11 describes cross-program coordination, alert protocols, and joint statistics. Section 12 presents costing and timeline. Section 13 assembles the decision tree through 2040. Section 15 concludes.

2 The target and the constraint landscape

2.1 Adopted parameters

Table 1 collects the parameters adopted throughout. We use the oMEGACat kinematic distance of 5.49 ± 0.06 kpc (Häberle et al., 2025), consistent within systematics with the Gaia EDR3 parallax distance (Soltis et al., 2021) and the combined-catalogue value (Baumgardt and Vasiliev, 2021). At this distance $1''$ subtends 0.0266 pc, so the entire fast-star region ($r < 3''$) fits within 0.08 pc, and the cluster core ($r_c \approx 2.4'$) within ~4 pc.

2.2 The mass tension

Table 2 and Figure 1 summarize the central-mass constraints. Two features matter for campaign design. First, the disagreement is not marginal: the velocity-only kinematic lower bound (8,200 $M_\odot$) exceeds the pulsar-timing point-mass upper bound (6,000 $M_\odot$, 3σ) outright, and the acceleration-informed lower bound (21,100 $M_\odot$) exceeds it by a factor of 3.5. At least one analysis is wrong, or incomplete, in an instructive way: for instance through the radial distribution assumed for the MSPs, the foreground/membership treatment of the fast stars, or the possibility that the central mass is genuinely extended (Bañares-Hernández et al., 2025; Zocchi et al., 2019; Breen and Heggie, 2013). Second, the tension is resolvable on a known schedule: more pulsars and longer timing baselines tighten the upper bound as roughly $1/\sqrt{N_{\rm psr} T^2}$ (Section 4); Roman and Gaia DR4 sharpen the astrometric frame within two years (Section 5); and LISA, if a compact-object inspiral is caught, measures the mass to ~ 0.1 per cent (Babak et al., 2017).

Table 1: Adopted parameters for ω Centauri (NGC 5139).

Quantity	Value	Source
Distance	5.49 ± 0.06 kpc	Häberle et al. (2025)
Cluster mass	$\approx 4 \times 10^6 \, M_\odot$	Harris (1996); Baumgardt and Vasiliev (2021)
Stellar count	$\sim 10^7$	Harris (1996)
Mean stellar age	12.08 Gyr (0.75 Gyr spread; systematics dominate)	Clontz et al. (2024)
Origin	stripped dwarf-galaxy nucleus	Hilker and Richtler (2000); Ibata et al. (2019); Souza et al. (2026)
Kinematic centre (J2000)	RA $13^{\mathrm{h}}26^{\mathrm{m}}47.24^{\mathrm{s}}$, Dec $-47°28'46.5''$ ($\pm 1''$)	Anderson and van der Marel (2010)
Known millisecond pulsars	19	Dai et al. (2020); Chen et al. (2023); Colom i Bernadich et al. (2026)
Central γ-ray source	4FGL J1326.7−4729 (MSP ensemble)	Dai et al. (2020, 2023)
Angular scale	$1'' = 0.0266$ pc	—

Table 2: Current constraints on the central dark mass of ω Cen (cf. Paper A, Table 2).

Constraint	Value	Method / source
Lower bound (velocities only)	$\geq 8{,}200 \, M_\odot$	7 fast stars, HST proper motions (Häberle et al., 2024b)
Lower bound (with accelerations, acceleration-model-informed)	$\geq 21{,}100 \, M_\odot$ (99%)	same stars, acceleration limits (Häberle et al., 2024b)
N-body growth models	~ 4.7–$5.1 \times 10^4 \, M_\odot$	cluster-evolution simulations (González Prieto et al., 2025)
Upper bound (point mass)	$< 6 \times 10^3 \, M_\odot$ (3σ)	stellar kinematics + MSP timing (Bañares-Hernández et al., 2025)
Favoured alternative	extended 2–$3 \times 10^5 \, M_\odot$	same analysis (Bañares-Hernández et al., 2025)
Timing (joint MeerKAT–Parkes)	insensitive at 10^3–$10^4 \, M_\odot$; $< 10^5 \, M_\odot$ (90%)	MSP timing (Colom i Bernadich et al., 2026)
Confirmed stellar-mass BH	$4.46^{+1.22}_{-1.01} \, M_\odot$ (94-yr binary, $a = 31$ AU)	HST/JWST astrometry (Whitaker et al., 2026)
Historical upper limits	$\lesssim 10^4 \, M_\odot$ (model-dep.)	HST proper motions (van der Marel and Anderson, 2010); see also Noyola et al. (2008); Zocchi et al. (2019)

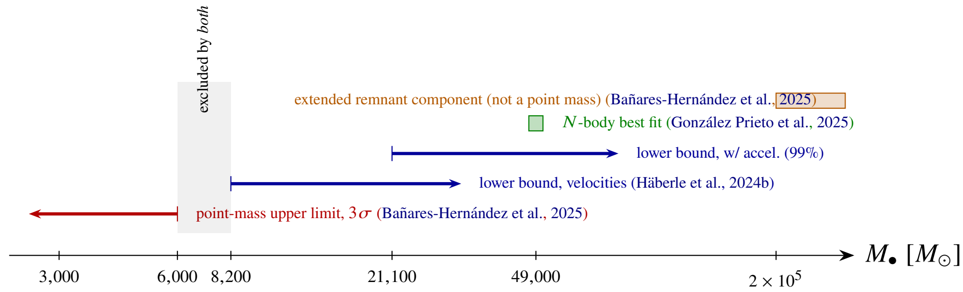

Figure 1: The central-mass constraint landscape (logarithmic mass axis). The pulsar-timing point-mass upper limit and the fast-star kinematic lower bounds leave no allowed point mass: at least one constraint must yield. The exclusions are not symmetric: the timing limit applies to a *point mass*, while the fast-star bounds apply to *any* bound central mass interior to the fast-star orbits, so an extended remnant component evades the former but not the latter. The campaign is structured to determine which constraint yields, on a known schedule, using instruments that share no dominant systematics.

2.3 Three hypotheses, fixed in advance

Following Paper A we carry three competing hypotheses through every program:

H_0 **(gas-starved IMBH or near-IMBH).** A genuine central point mass in the $\sim 10^4\,M_\odot$ range exists and is silent for the ordinary reason: a relaxed, gas-poor, 12-Gyr-old cluster supplies essentially nothing to accrete. Recent survival analyses find that IMBHs formed early in dense clusters are plausibly retained to the present day in ω Cen-like hosts, supporting the prior on this branch (Martinez et al., 2026). Default and most parsimonious if the kinematic bounds hold.

H_1 **(extended remnant component).** No IMBH; the central mass is a spatially extended cluster of stellar remnants (Bañares-Hernández et al., 2025; Breen and Heggie, 2013; Zocchi et al., 2019). Default and most parsimonious if the timing bound holds.

H_2 **(anomalous/managed system).** A central IMBH whose observational properties are shaped by technology, per Paper A. Carries a low prior; becomes interesting only through conjunctions that H_0 and H_1 jointly fail to explain (e.g. near-extremal spin on a gas-starved hole; repeated coincident multi-messenger bursts; statistically anomalous depletion of the loosely bound core population).

The campaign's purpose is to measure the system well enough that H_0 and H_1 are decided on their merits, with H_2's distinctive residues either appearing in the data or being retired by them, rather than to hunt for H_2. These labels map one-to-one onto the hypothesis set of the companion adjudication framework (Swanson, 2026a, hereafter Paper E): H_0 corresponds to Paper E's H_q (quiescent astrophysical IMBH), H_1 to $H_{\rm sub}$ (sub-IMBH remnant alternative), and H_2 to $H_{\rm eng}$ (engineered system).

3 Program 1: JWST infrared accretion and waste-heat limits

Bottom line. Deep two-epoch JWST imaging of the core is the decisive observable: an accretion-like point source, or a mid-infrared excess over the stellar-population SED carrying the warm/cool swarm colour discriminant. The current best limits are archival: radiative output $\lesssim 10^{-9}$ of Eddington, radio accretion $\lesssim 4 \times 10^{-3}$ of the Bondi rate, and X-ray $L_X \lesssim 1.6 \times 10^{30}$ erg s^{-1}. The target sensitivity is ~5–10 nJy (3σ) at F444W, a bolometric limit of order 10^{30} erg s^{-1}, and a tightened ceiling of $P_{\rm comp} \lesssim 10^3$–$10^4\,L_\odot$ on managed throughput. The result feeds the deepest multi-band quiescence statement available before LISA, and any excess or variability enters the coordination protocol of Section 11.

3.1 Science case

For any central mass $M_\bullet$, the Eddington luminosity is $L_{\rm Edd} \simeq 1.26 \times 10^{31}\ (M_\bullet/M_\odot)$ W: 1.0×10^{35} W at 8,200 $M_\odot$, 2.5×10^{35} W at $2 \times 10^4\ M_\odot$, 6.2×10^{35} W at $4.9 \times 10^4\ M_\odot$. The existing non-detections already establish extreme quiescence: the radio limit bounds accretion at $\lesssim 4 \times 10^{-3}$ of the Bondi rate (Mahida et al., 2026), the archival Chandra limit bounds the X-ray output at $L_X \lesssim 1.6 \times 10^{30}$ erg s^{-1} (Haggard et al., 2013), and the JWST photometry bounds the radiative output at $\lesssim 10^{-9}$ of Eddington (Chen et al., 2026), among the most extreme quiescence levels measured for any black hole. The published JWST analysis, however, is an archival by-product, not an optimized deep program. A purpose-designed campaign improves the limits by an order of magnitude and adds the two dimensions the archival data lack: variability (accretion at these rates is expected to flicker; a managed or artificial suppression of accretion, the H_2 residue, is not) and waste heat (a mid-infrared excess over the stellar population model is the classic Dyson-type technosignature; Dyson 1960; Wright et al. 2014; Hsiao et al. 2021, and at ω Cen the relevant solid angle is a few arcseconds, not a galaxy).

3.2 Current and scheduled JWST programs

Two approved JWST programs already execute pieces of this campaign, and the deliverables below are stated as increments over them. GO 5137 (PI Seth), "Weighing the Intermediate Mass Black Hole in Omega Centauri," obtains a deep NIRSpec IFU mosaic of the central field: line-of-sight (LOS) velocities for ~70 stars in the innermost region, including all four fast movers inside 1″, together with the deepest search to date for accretion-emission signatures at the candidate position (Seth et al., 2024). GO 8322 extends the astrometric time base with dedicated JWST monitoring of the fast stars (JWST GO 8322 team, 2026). Three consequences for Program 1: (i) the accretion-emission search is no longer virgin territory — the deep-imaging deliverable is the order-of-magnitude *photometric* depth gain, the second epoch for variability, and the 10–25.5 μm waste-heat axis (the long-wavelength limit of MIRI imaging), none of which the IFU program provides; (ii) the contingent spectroscopic environment test (below) becomes an extension of GO 5137's footprint rather than a first observation; and (iii) the LOS velocities feed directly into the astrometric program, where they convert two-dimensional proper motions into three-dimensional velocity vectors and open a radial-velocity acceleration channel (Section 5.1).

3.3 Implementation

Deep imaging (new observations). NIRCam F200W/F356W/F444W at 20 ks per filter plus MIRI F770W/F1000W at 15 ks per filter, in two epochs separated by ~6 months: ~65 h total. In the crowded core the limits are confusion- rather than photon-limited. The NIRCam figures are ETC-class point-source depths scaled to the requested 20 ks exposure and degraded by an assumed ~2.9× crowding penalty, reaching ~5–10 nJy (3σ) across F200W/F356W/F444W; the empirical-PSF-subtraction gain, anchored to the oMEGACat astrometric catalogue (Häberle et al., 2024a; Nitschai et al., 2023), is named as an assumption to be verified by scene simulation, not a measured recovery. Chen et al.'s own archival NIRCam photometry of this field reaches 660 nJy–1.6 μJy at their 515 s exposure (Chen et al., 2026), which implies a crowding penalty of ~29–70× without PSF subtraction and puts the adopted ~2.9× assumption in view as an assumption rather than a demonstrated anchor. At MIRI the ETC-class point-source depths at 15 ks are ≈70 nJy (3σ) at F770W and ≈170 nJy at F1000W before crowding, degrading to ≈200 and ≈500 nJy respectively after the crowding penalty discussed next. At the ω Cen core stellar density ($\rho_0 \simeq 3 \times 10^3\, M_\odot\, \mathrm{pc}^{-3}$; $\sim 10^7$ stars in the field) the NIRCam F444W pixel scale (0.063″) resolves of order fifty stars per resolution element in the inner arcsecond, and the 90 per cent injection depth is a property of the local surface density and the PSF-fitting algorithm rather than the instrument. Table 3 breaks the scalar ∼3 crowding factor into per-filter completeness versus photon limits, using the same Chen et al. 90 per cent completeness curves that set the Paper F infrared leg (their Table 1, scaled by exposure time; Paper F Appendix 14 and `miri_sensitivity.json`, ETC 6.0). For each filter the ETC photon-limited 3σ depth at the requested exposure and the 90% injection-recovery depth are compared; the adopted depth is the shallower. At the requested 20 ks (NIRCam) and 15 ks (MIRI) the completeness depth is shallower in every filter, so the program is crowding-limited and its exposure is sized to that wall (Table 3). Doubling the integration improves the photon limit by $1/\sqrt{2}$ but moves the adopted limit by <0.1 dex, while halving it would make sensitivity bind. This pre-empts the TAC question of why 15 ks rather than 8 ks at MIRI: at 8 ks the photon and completeness depths are comparable, and at 15 ks the photon depth sits a factor ∼1.5 below the completeness wall, so the time request buys the wall rather than empty depth. The MIRI figures carry the larger systematic: RGB-tip giants are mJy-level sources at 10 μm whose extended PSF halos overlap in the core, in-flight long-wave throughput degradation must still be budgeted, and the quoted depths assume the NIRCam-calibrated ∼3× penalty transfers to the mid-IR PSF—assumptions a full scene simulation must verify before the time request is finalized.

Table 3: Per-filter depth at the requested exposure: photon limit versus 90% completeness (Chen et al. 2025, as used in Paper F) and which binds. MIRI ETC from `miri_sensitivity.json` (ETC 6.0, 10 ksec scaled as $t^{-1/2}$; rescaled 61 nJy at F770W and 115 nJy at F1000W 3σ, printed as ≈70 and ≈170 nJy before crowding); NIRCam ETC/completeness from the oMEGACat F444W injection that recovers 90% at the quoted depth, scaled by exposure and pixel scale with F356W interpolated. Adopted is the shallower; all five filters are crowding-limited at the requested depth, so exposure is sized to the completeness wall.

Filter	$\lambda_{\rm piv}$ (μm)	Exp. (ks)	ETC photon 3σ (nJy)	90% completeness (nJy)	Binds	Adopted 3σ (nJy)
F200W	2.00	20	4	12	crowding	12
F356W	3.56	20	3	9	crowding	9
F444W	4.44	20	3	8	crowding	10[a]
F770W	7.70	15	61 (70[b])	184 (200[b])	crowding	200
F1000W	10.0	15	115 (170[b])	345 (500[b])	crowding	500

[a] Printed as ∼5–10 nJy (3σ) across F200W/F356W/F444W, anchored to the injection; adopted 10 nJy is the shallower end, consistent with 3× penalty. [b] Parentheses are the rounded values printed in the pre-table sentence; first numbers are the rescaled ETC/completeness from `fC_calc7_miri.json`.

At 5.49 kpc, 10 nJy at F444W corresponds to a monochromatic $\nu L_\nu \approx 2.5 \times 10^{28}$ erg s^{-1}; for plausible quiescent-accretion SEDs the bolometric limit is of order 10^{29}–10^{30} erg s^{-1}, i.e. Eddington ratios of 4×10^{-14}–4×10^{-13} at $2 \times 10^4\, M_\odot$ — probing a regime in which even Sgr A* would be conspicuous (Event Horizon Telescope Collaboration, 2022).

Two-epoch variability. Differential photometry between epochs is robust to the static crowding systematics that dominate the absolute limits; variability at the ≥20–30 per cent level for any source above ∼30 nJy at NIRCam (above ∼0.5 μJy at F1000W) flags an accretion candidate and triggers the coordination protocol of Section 11. Two-epoch photometry should not be mistaken for a test of Paper E's regulation statistic R, the variability *deficit* in the MAD flux-eruption band (10^{-3}–10^{-1} Hz for a $2 \times 10^4\, M_\odot$ hole; cycle times of 10–100 s), which requires fast-cadence monitoring. At the fiducial mass that is an X-ray measurement, not a radio one: a compact self-absorbed synchrotron source low-pass filters the eruption band at the source, and interstellar scintillation adds variability uncorrelated with accretion on top, so the radio band carries no discriminating power for R; the channel activates at $L_X \gtrsim 10^{34}$–10^{35} erg s^{-1}, i.e. in flare or TDE states (Swanson, 2026a). Accordingly, if accretion is ever detected at that level in any band, a fast-cadence X-ray contingency activates (pointed timing plus the archival X-ray channel of Program 8), targeted at the eruption band, with Program-2 radio monitoring serving as context rather than as an R measurement.

Waste-heat analysis (archival + new). The 10–25.5 μm field photometry is fitted against the synthetic stellar-population SED constructed from the oMEGACat spectroscopic catalogue (Nitschai et al., 2023). The primary waste-heat deliverable is a *point-source* SED excess with a colour discriminant: the configurations surviving Paper E's feasibility analysis are cool swarms at 10^2–10^3 AU, which subtend $0.018''$–$0.18''$ at 5.49 kpc and are unresolved inside the F1000W PSF core ($0.33''$), so morphology carries no information inside that envelope and the discriminant is spectral: a 50–300 K near-blackbody with no silicate features, against the AGB dust shells that dominate the false-positive population (Swanson, 2026a). Paper E's Appendix B forward model, a two-temperature swarm SED integrated against the nine MIRI imaging filters, supplies this discriminant quantitatively: any point-source excess is tested against the predicted warm/cool flux ratio across F560W–F2550W rather than a single-band flux, which any AGB dust shell of comparable brightness will not reproduce over the same filter set. We retain the extended statistic (a spatially coherent excess in the central 0.1 pc, i.e. $3.8''$, exceeding 3σ over the population model) but relabel it: it is a hypothesis-agnostic anomaly channel, no longer the test of the surviving waste-heat configurations. The deliverable is quantitative through Paper E's waste-heat floor, $L_{\mathrm{waste}} \geq 10^{-4}\, P_{\mathrm{comp}}$: the current JWST-derived limit converts to a ceiling on sustained computational power of $P_{\mathrm{comp}} \lesssim 10^3$–$10^4\, L_\odot$, and the ceiling tightens linearly with mid-infrared depth (Swanson, 2026a). One temperature qualifier bounds every MIRI-derived ceiling: a 50 K swarm peaks near 60 μm and is essentially invisible at ≤ 25.5 μm, so the ceiling applies to the warm ($\gtrsim 150$ K) end of the configuration space only, and the cold end is unconstrained by any instrument in this campaign. The same fit yields conventional science: the most precise mid-IR census of the post-main-sequence population in any globular cluster.

Spectroscopic environment test (Cycle 6+, contingent). The approved GO 5137 IFU observations (Section 3.2) already cover the innermost arcseconds; the contingent extension proposed here is a wider NIRSpec IFU mosaic of the inner 0.5 pc (~125 h), activated only if astrometry or timing strengthens the IMBH case, mapping abundance anomalies (Li enhancement, refractory depletion) that would discriminate between ordinary tidal-disruption debris chemistry (Hills, 1988; Gezari, 2021) and the engineered-feeding residue of H_2. We do not request this time until the trigger condition is met.

3.4 Decision criteria

Null result (expected under H_0, H_1, and mature-H_2 alike): bolometric limit $\lesssim 10^{30}$ erg s^{-1} recorded (radiative output $\lesssim 10^{-12}$ of Eddington across the contested mass range), complementing the radio Bondi-efficiency bound and jointly constraining any radiatively inefficient flow via the fundamental plane (Merloni et al., 2003; Mahida et al., 2026). A static detection consistent with a quiescent accretion SED supports H_0 and immediately sharpens every dynamical program (the source position becomes the kinematic centre). A variable detection triggers multi-messenger follow-up. A $> 3\sigma$ mid-IR excess (point-source with the swarm colour discriminant, or extended with no stellar counterpart) is the single strongest electromagnetic anomaly the campaign can produce and would be adjudicated under Section 11 rules.

4 Program 2: radio deep imaging, narrowband SETI, and pulsar timing

Bottom line. The decisive observable is the radial profile of millisecond-pulsar accelerations across the cluster, which distinguishes a central point mass from an extended remnant component; the narrowband SETI limit and the deeper continuum image ride on the same sessions. The current best limits are the 3σ point-mass cap of $\sim 6 \times 10^3\, M_\odot$ from the joint kinematic-timing analysis and the latest joint MeerKAT–Parkes insensitivity at 10^3–$10^4\, M_\odot$ with a 90-per-cent ceiling of $10^5\, M_\odot$. The target is ten well-timed pulsars on ten-year baselines, pushing the bound to the 3,000–4,000 $M_\odot$ level or finding the profile, plus a first ω Cen technosignature limit near 4×10^{15} W EIRP over the full cluster. This program feeds decision point D2 directly.

4.1 Science case

The radio program carries three loads at once. (i) *Continuum*: a deeper interferometric limit (or first detection) on the central compact source, extending Mahida et al. (2026) and the globular-cluster IMBH radio campaigns (Strader et al., 2012; Tremou et al., 2018). (ii) *Narrowband SETI*: to our knowledge the first dedicated technosignature search of any kind at ω Cen, closing the gap left by the FAST pilot survey's latitude limit (Huang et al., 2026b). (iii) *Pulsar timing*: the binding constraint on the central mass comes from the cluster's MSP population (five discovered with Parkes, Dai et al. 2020, timed to $\dot{\nu}$ within 3.5 yr, Dai et al. 2023; then thirteen more with MeerKAT/TRAPUM, Chen et al. 2023; Stappers and Kramer 2016). The joint MeerKAT–Parkes timing analysis now extends both the census and the constraints: it reports a nineteenth discovery, PSR J1326−4728S, bringing the known population to nineteen, and its measurements are insensitive to an IMBH of 10^3–$10^4\, M_\odot$ while placing a 90 per cent upper limit of $< 10^5\, M_\odot$ on the central mass (Colom i Bernadich et al., 2026). The same analysis finds an anomalously high fraction of isolated and black-widow systems among the cluster's MSPs, which bears directly on the MSP radial-distribution priors on which the Bañares-Hernández et al. (2025) timing bound leans, since the spatial distribution of a dynamically processed population need not follow the assumed profile. The path to resolving the mass tension runs directly through more pulsars and longer baselines (Bañares-Hernández et al., 2025; Chen et al., 2025).

4.2 Implementation: MeerKAT era (2026–2030)

Timing. Bi-weekly L-band sessions (~26 per year, ~3–5 h each) on all nineteen MSPs with MeerKAT (Jonas and MeerKAT Team, 2016), producing times of arrival at several-μs precision for the brightest objects (the cluster's MSPs are 10–30 μJy sources; sub-μs timing is not on offer). Line-of-sight acceleration precision from timing scales steeply with baseline ($\sigma_a \propto T^{-5/2}$ for white noise); by analogy with the mature 47 Tuc and Terzan 5 programs (Freire et al., 2017; Prager et al., 2017), five years of MeerKAT data yield per-pulsar acceleration uncertainties of order 10^{-10} m s^{-2} for the best-timed objects. We do not extrapolate to the 10^{-11} decade: the 47 Tuc and Terzan 5 precedents rest on 20–25-yr baselines, and DM variation and red timing noise break the white-noise $T^{-5/2}$ scaling before that precision is reached (van Haasteren and Levin, 2013). For scale, a $4 \times 10^4\, M_\odot$ point mass imposes $a \approx 6 \times 10^{-8}$ m s^{-2} at $r = 0.3$ pc: the discriminating signal is not the detection of acceleration (already achieved; Dai et al. 2023; Bañares-Hernández et al. 2025) but the radial profile of accelerations and jerks across the MSP population, which distinguishes a point mass ($a \propto r^{-2}$) from an extended remnant component. Commensal search observations are expected to add 5–10 MSPs (the luminosity function is far from exhausted; Chen et al. 2023), and the population-level point-mass upper bound tightens approximately as $1/\sqrt{N_{\rm eff}}\, (T_0/T)^{\geq 1}$; this is an ensemble, bound-level scaling, distinct from the per-pulsar white-noise acceleration scaling

$\sigma_a \propto T^{-5/2}$ quoted above, because the ensemble constraint is limited by the pulsars' unknown line-of-sight positions and intrinsic spin-down as well as by timing noise. The effective sample size is not 19. The released timing record (`paper/h/data/pulsars.json`) carries published timing solutions with 4–5.1-yr baselines for twelve of the nineteen pulsars — six others have no timing solution in the record, and one more is a single-epoch discovery — and the per-pulsar acceleration sensitivities, $\sigma_{a,i} \propto P_i\, \sigma_{t,i}/T_i^2$, differ through the spin periods (2.5–19 ms), so the inverse-quadrature sum with weights $w_i = 1/\sigma_{a,i}$ gives $N_{\rm eff} = (\sum_i w_i)^2/\sum_i w_i^2 = 9.8$. The ensemble sensitivity is dominated by the twelve pulsars with published timing solutions and multi-year baselines; the timing residual enters $\sigma_{a,i}$ as a common factor (several-μs ToA precision, not sub-μs) and cancels in $N_{\rm eff}$. Ten well-timed pulsars with ten-year baselines push a 6,000 $M_\odot$ bound to the 3,000–4,000 $M_\odot$ level if no point-mass signal emerges, or find one.

Forecast requirement (pre-registered). The profile discriminator just described is the campaign's primary near-term test, and it is at present asserted rather than demonstrated: most of the MSPs sit well outside the central 0.1 pc, and their unknown line-of-sight positions degrade the radial information the test needs. Before the D2 data lock (decision point D2 of the D1–D4 sequence defined in Section 13) we will therefore complete and publish a mock-population forecast: point-mass and extended-component potentials injected at the real MSP sky positions, line-of-sight positions marginalized over the cluster density profile, and realistic timing noise applied, reporting the recoverable Bayes factor (or $\Delta\chi^2$) as a function of baseline and census size. An order-of-magnitude expectation frames the exercise: with 19–25 MSPs, 5–10-yr baselines, and per-pulsar acceleration uncertainties of order 10^{-10} m s^{-2}, the profile difference between a $2 \times 10^4\, M_\odot$ point mass and a 2–$3 \times 10^5\, M_\odot$ extended component is of order 10^{-9}–10^{-8} m s^{-2} for the innermost pulsars at their favourable, near-plane-of-sky assumed 3D positions, suggesting $\Delta\chi^2$ of order tens ($\ln K \sim 5$–15) in that favourable case and degrading toward indistinguishability once the true line-of-sight positions are marginalized over the cluster density profile, as the pre-registered forecast below does; the forecast determines where between those poles the real configuration falls. Its pre-registered success criterion is a median discrimination of $\Delta\chi^2 \geq 9$ between the two potentials at the D2 epoch; if the forecast falls short, the D2 decision weight shifts to the astrometric channels of Section 5. A complementary dynamical diagnostic is available from the same data: an IMBH measurably alters the degree of energy equipartition of the host cluster (Aros and Vesperini, 2023), so the equipartition profile inferred from the kinematic catalogues provides an independent axis along which the point-mass and extended hypotheses separate.

Narrowband SETI. Commensally with every timing session, a 1-Hz-resolution spectrometer backend records the full primary beam (FWHM $\sim 1^\circ$ across most of L band, containing the whole cluster). Processing follows standard drift-search practice (Enriquez et al., 2017): Doppler drifts of ± 4 Hz s^{-1} at L band (a fractional convention, $\dot{\nu}/\nu \approx 3 \times 10^{-9}$ s^{-1}; the absolute window scales with observing frequency for any S-band search), on–off cadence against reference pointings, and a candidate-advance threshold of SNR > 15. The sensitivity limits below are quoted at the conventional SNR = 10 of Eq. (1); the two numbers serve different roles (limit-setting versus pipeline promotion) and are stated separately to keep them from being conflated. Seventy hours over two years reach a minimum detectable EIRP of $\sim 4 \times 10^{15}$ W at 5.49 kpc. That threshold follows from the standard narrowband relation

$$\mathrm{EIRP}_{\min} \;=\; 4\pi d^2\, \mathrm{SNR}\ \mathrm{SEFD}\,\sqrt{\delta\nu/t}, \tag{1}$$

with $d = 5.49$ kpc, SNR = 10, $\delta\nu = 1$ Hz, and $t = 70$ h. The SEFD entering Eq. (1) is an array figure, not an antenna figure: the ≈ 430 Jy value is the per-antenna SEFD of a single 13.5-m MeerKAT dish (Jonas and MeerKAT Team, 2016), and an $N \approx 60$-antenna sum improves on it by $\sqrt{N}$ to ≈ 55 Jy incoherently, or by N to ≈ 7 Jy in a coherent tied-array beam. The quoted 4×10^{15} W is the incoherent, full-cluster figure; a coherent tied-array pointing reaches $\approx 5 \times 10^{14}$ W but covers only an arcseconds-scale beam.

These limits are quoted within the half-light radius: the EIRP limit degrades by 2× at the half-power radius of the primary beam, a transmitter at the ∼0.8–1° tidal radius (Harris catalogue 1.06°, Harris 1996; a Nature-paper King-profile fit gives 0.81°, Häberle et al. 2024b) sits at or beyond the half-power point at the top of L band, and the S-band primary beam ($\sim 40'$) does not contain the cluster at all. The incoherent full-cluster limit is below the FAST pilot's $\sim 10^{16}$ W thresholds at its closer, northern targets (Huang et al., 2026b); it is the first limit of any kind for this cluster, and it is sufficient to detect any transmitter exceeding $\sim 2 \times 10^2$ times the EIRP of the Arecibo planetary radar ($\sim 2 \times 10^{13}$ W; Enriquez et al. 2017, a modest output for an energy-rich civilization) from any of $\sim 10^7$ stars in the beam simultaneously. For comparison with FAST's per-target accounting, a single $5''$ coherent follow-up beam at the core contains $\sim 10^3$ stars, so even the deep coherent limit is a per-thousand-star figure rather than a per-star one.

Backend requirement: spatial RFI discrimination. An incoherent full-primary-beam sum has no spatial discrimination: every signal within $\sim 1°$ enters the data product, and there is no on/off inside a single pointing. The dominant contaminant for a 2026+ campaign is not fixed-frequency terrestrial RFI but LEO mega-constellation downlinks sweeping through L band with Doppler drifts inside the searched ± 4 Hz s^{-1} window; candidate rates of 10^4–10^5 hr^{-1} are routine in current data. The mitigation is architectural rather than procedural: simultaneous coherent beams with candidate cross-rejection, the BLUSE architecture of Czech et al. (2021), in which a candidate appearing in widely separated beams is rejected as local. Tiling the half-light region with $5''$ coherent beams would require thousands of beams; the realistic design forms a modest set of anti-coincidence beams plus a reduced-baseline subarray whose wider coherent beams cover the core at intermediate SEFD. We state this as a hard backend requirement, because without it the false-candidate budget, not the radiometer equation, sets the program's labour cost.

Scintillation-aware confirmation. The 5.49-kpc, DM ≈ 100 pc cm^{-3} sightline scintillates strongly at L band: diffractive scintillation modulates a narrowband signal by factors of several on minutes-to-hours timescales, the case treated for SETI by Cordes et al. (1997). A criterion requiring re-detection in a single independent session would therefore reject a genuine steady transmitter with order-unity probability whenever the confirming epoch lands in a scintillation null. The pre-registered confirmation policy is scintillation-aware: a candidate surviving the cross-rejection screen is scheduled for multiple re-observation epochs spaced by at least the refractive timescale, and confirmation significance is computed under exponential-intensity statistics rather than a constant-flux assumption, so that a non-detection in any single epoch demotes rather than kills the candidate.

Continuum. The summed continuum visibilities from the timing campaign provide a ∼100+ h synthesis dataset at L/S band, and the depth budget is set by confusion, not thermal noise: at the $\sim 6''$ resolution of a naturally weighted L-band image the confusion limit is formalism-dependent: the classical one-source-per-beam criterion gives ≈ 0.35 μJy, while the Condon (1974) self-consistent criterion (cutoff at five times the rms confusion; Condon 2007) gives ≈ 3.5 μJy, against a thermal noise of 0.41 μJy in 100 h (Table 4). The confusion-versus-thermal ordering at $6''$ therefore depends on the formalism, while at UHF confusion dominates in both; 19+ steep-spectrum MSPs sit at the position of interest either way. The useful deliverables are therefore long-baseline, uniform-weighted imaging of the central arcminute (where the confusion limit is far deeper), UHF/S-band spectral decomposition separating the steep-spectrum pulsar population from any flat-spectrum accretion candidate, and in-beam variability of compact sources; combined with the ATCA 7.25-GHz limit (Mahida et al., 2026) these constrain both the flat-spectrum (jet) and steep-spectrum (pulsar) interpretations of any future central source candidate.

4.3 Implementation: SKA era (2029–)

SKA-Mid science verification is expected from 2027, shared-risk operations from 2029, and early full operational cycles in the early 2030s (Braun et al., 2019). The ω Cen program transfers wholesale:

timing precision improves by the sensitivity ratio (factor ~4–5 over MeerKAT for these declinations), the MSP census plausibly doubles, and the narrowband EIRP thresholds drop by the same factor, toward $\sim 10^{15}$ W incoherent and $\sim 10^{14}$ W coherent, on a southern target FAST cannot see. (The ngVLA, the other next-decade radio flagship, is not an option here: at $\delta = -47.5°$ ω Cen is inaccessible from its planned sites.) By LISA launch, the pulsar acceleration map will be the best non-GW constraint on the central potential in any globular cluster.

4.4 Decision criteria

The timing program is the campaign's primary near-term discriminator between H_0 and H_1: a smooth r^{-2} acceleration profile centred on the kinematic centre at $\geq 10^4\,M_\odot$ retires H_1; a persistently tightening upper bound below the kinematic lower bounds forces revision of the fast-star analysis (membership, foreground, or binarity systematics) and shifts the campaign's centre of gravity to H_1. For SETI: any re-detected narrowband candidate enters the two-messenger protocol; a null is published as the first ω Cen technosignature limit. Paper A dependence (labelled): under H_2, P3 of that paper predicts no leakage radiation, so SETI nulls carry no evidential weight against H_2; they constrain only conventional beacon scenarios. Consistent with this, Paper E's Bayes-factor analysis finds that additional radio depth beyond current limits moves $\ln K$ only weakly for every hypothesis pairing (Swanson, 2026a); the case for the deeper continuum limit therefore rests on the H_0/H_1 accretion physics, not on technosignature discrimination.

4.4.1 Confusion per band

Classical confusion estimates differ by about an order of magnitude at these resolutions depending on formalism, so Table 4 carries the two defensible readings per band rather than one number: the one-source-per-beam cutoff at half its flux, and the Condon self-consistent criterion, whose cutoff sits at five times the rms confusion (Condon, 2007); the framework is the flux-density error-distribution formalism of Condon (1974). Both readings use the same deep 1.4-GHz count law ($N(> S) \approx 220\,(S/\mathrm{mJy})^{-1}\,\mathrm{deg}^{-2}$ across 10 μJy–1 mJy, flattening toward the Euclidean slope above 1 mJy), whose normalization carries about a factor-of-two uncertainty; the band-to-band ratios are more robust than the absolute scale. The thermal column is the correlated-array radiometer noise of Section 4. Numbers in `figs/fC_calc7_confusion.json` and `figs/fC_calc7_condon_fix.json`.

Table 4: Confusion (μJy beam^{-1}) under two formalisms, against the 100-h thermal noise. The Condon column applies the criterion of Condon (2007): the cutoff is set at five times the rms confusion.

Band	ν / θ	1-per-beam	Condon $q = 5$	thermal
L	1.4 GHz / 6″	0.35	3.46	0.41
S	3.0 GHz / 6″	0.35	2.03	—
UHF	0.8 GHz / 8″	0.62	9.11	—
L, uniform	1.4 GHz / 4″	0.15	1.54	—

5 Program 3: astrometry with HST, Gaia DR4, Roman, and ELT/MICADO

Bottom line. The single decisive observable is the Gaia DR4 re-derivation of the fast-star velocity excess on the improved absolute frame, the cheapest measurement in the campaign that can either firm the mass tension into a contradiction or dissolve it; Roman photocentric wander and MICADO inner-star discovery carry the longer-term mass discrimination. The current best astrometry is the oMEGACat catalogue, against which direct acceleration detection is below 1σ before $\sim$2040. The targets are the DR4 proper-motion improvement (several-fold over DR2-era ties), few-μas-class wander measurement from Roman, and new fast stars at $r < 0.02$ pc from MICADO. The DR4 re-derivation feeds decision point D1 in 2027; wander and inner stars feed D2 and the LISA prior.

5.1 Sensitivity of the direct acceleration test

The intuitively obvious astrometric test (watch the seven fast stars curve) does not work on a useful timescale, which we now demonstrate quantitatively. The proper-motion acceleration induced by a central mass $M_\bullet$ at projected radius r is

$$a \;=\; \frac{GM_\bullet}{r^2} \;\approx\; 4.4\times10^{-7}\left(\frac{M_\bullet}{2\times10^4\,M_\odot}\right)\left(\frac{r}{0.08\,\mathrm{pc}}\right)^{-2}\,\mathrm{m\,s^{-2}}, \tag{2}$$

which at $d = 5.49$ kpc corresponds to $\approx 5.3\times10^{-4}$ mas yr^{-2} (mean projection factors of order unity absorbed). Against this, the oMEGACat astrometry, whose released proper-motion precisions over its $\sim$20-year HST baseline imply an effective per-epoch positional precision of order 0.02 mas rather than a directly reported single-epoch quantity (Häberle et al., 2024b,a), delivers acceleration uncertainties of $\sigma_a \approx 1\times10^{-3}$ mas yr^{-2} per star: a 0.5σ measurement. Extending HST monitoring to 2028 (a 26-year baseline) improves this only to $\sim 0.7\sigma$; a four-year ELT/MICADO campaign at $\sim$50–100 μas per epoch (Davies et al., 2021), run through the least-squares $\sqrt{N}$ chain $\sigma_a = 26.83\,\sigma_{\mathrm{pos}}/(\sqrt{N}\,T^2)$ at 2 epochs yr^{-1} ($N = 2T$; the estimator used throughout this section and in Figure 2), reaches $\sigma_a \approx (3.0$–$5.9)\times10^{-2}$ mas yr^{-2} (4×10^{-2} at the fiducial 70 μas): worse, because acceleration precision scales as $T^{-2.5}$ under fixed-cadence monitoring and four years is short (Figure 2). Direct 5σ curvature detection at nominal parameters requires either $M_\bullet \gtrsim 10^5\,M_\odot$, a star at $r < 0.015$ pc, or $\gtrsim$30-year baselines on ELT-class astrometry (the same chain gives 38–44 yr at 50–70 μas). We therefore retain the fast-star monitoring (cheap; protects against the lucky cases, since González Prieto et al. 2025-mass holes and undiscovered inner stars are both live possibilities) but route the decision-grade astrometry through the three astrometric measurements below.

The approved GO 5137 spectroscopy (Section 3.2) adds two levers this budget did not include. First, the LOS velocities of the fast movers convert their two-dimensional proper motions into three-dimensional velocity vectors; since the LOS component can only increase a star's total speed, each measurement can only tighten the escape-velocity lower bound of Häberle et al. (2024b), star by star. Second, repeat NIRSpec visits open a radial-velocity-drift acceleration channel independent of the proper-motion curvature budget above. The expected LOS acceleration at the fiducial parameters of Eq. (2) is $a_{\mathrm{los}} \approx 4.4\times10^{-7}$ m s^{-2} ≈ 0.014 km s^{-1} yr^{-1}. Assuming per-epoch RV precision of $\sim$3 km s^{-1} (crowded-field NIRSpec on faint members) and annual visits over five years with white noise, the recoverable drift uncertainty is $\sigma_{\dot v} \approx \sigma_{\mathrm{RV}}\sqrt{12}/(\sqrt{n}\,T) \approx 0.9$ km s^{-1} yr^{-1}: a factor $\sim$70 above the nominal signal, the same verdict as the proper-motion channel (Figure 2, lower panel). The channel is nonetheless worth running because it shares none of the proper-motion systematics (frame tie, geometric distortion) and inherits the full r^{-2} gain for any inner star: at $r = 0.01$ pc the drift signal reaches $\sim$0.9 km s^{-1} yr^{-1}, reaching parity ($\sim 1\sigma$) with $\sigma_{\dot v}$ within a five-year GO 5137 extension; a 3σ detection requires a $\sim$decade baseline or higher-cadence visits.

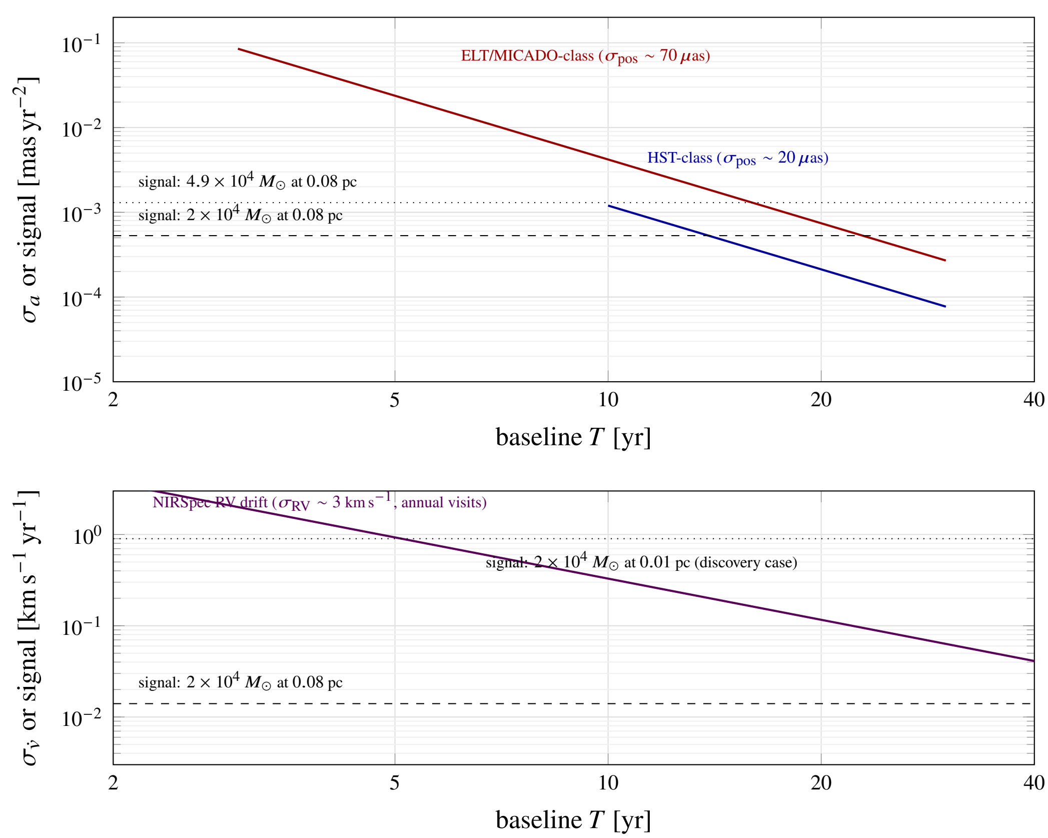

Figure 2: Why direct acceleration detection on the known fast stars is marginal. *Upper panel:* 1σ acceleration uncertainty versus baseline for HST-class and ELT-class astrometric campaigns (curves, the least-squares $\sqrt{N}$ chain $\sigma_a = 26.83\,\sigma_{\rm pos}/(\sqrt{N}\,T^2)$ at 2 epochs yr^{-1} ($N = 2T$), anchored to the measured oMEGACat performance ($\sigma_{\rm pos} = 20\,\mu$as) and nominal MICADO performance ($\sigma_{\rm pos} = 70\,\mu$as)), against the expected signals (horizontal lines, Eq. 2). Curves assume uniform observing cadence over the full baseline and are therefore optimistic for sparse epoch-limited extensions (the realistic 2028 HST extension yields only $\sim 0.7\sigma$). Under this scaling the HST curve crosses 5σ on the heavy González Prieto et al. (2025) mass at ∼18 yr and on the nominal mass at ∼26 yr; the ELT/MICADO curve crosses 5σ on the heavy mass only at ∼30 yr and does not reach 5σ on the nominal mass within the plotted baseline. *Lower panel:* the radial-velocity-drift channel opened by GO 5137-class NIRSpec monitoring (Section 5.1): at the nominal fast-star radius the verdict matches the proper-motion channel, but an inner star at 0.01 pc crosses the sensitivity curve within a few years. The astrometric program is therefore built around wander, reference-frame, and discovery measurements instead (Section 5.2).

5.2 Three astrometric measurements

(i) Photocentric wander (Roman). An IMBH of mass $M_\bullet$ in dynamical equilibrium with a stellar bath executes Brownian motion with velocity $\sigma_\bullet \sim \sigma_* \sqrt{m_{\rm eff}/M_\bullet}$ relative to the cluster centre of mass. The effective mass is the mass-weighted mean $m_{\rm eff} = \langle m^2\rangle/\langle m\rangle$, not the number-weighted $\bar{m}$, because the velocity diffusion scales with $n\langle m^2\rangle$ while the drag scales with $n\langle m\rangle$ (Chatterjee et al., 2002; Merritt et al., 2007). The distinction is worth a factor of a few here and it runs in the program's favour: for the segregated mass function adopted in Paper E, $m_{\rm eff} \simeq 2.3\,M_\odot$ against $\langle m\rangle = 0.54\,M_\odot$, so the amplitude is ~2.7× larger than an equal-mass estimate at $\bar{m} = 0.3\,M_\odot$ gives, ~13 μas yr^{-1} rather than ~5 at 8,200 $M_\odot$. The lighter the hole, the larger its wander: the predicted amplitude differs by ~17 per cent between 6,000 and 8,200 $M_\odot$ (the square root of the mass ratio), and by a factor ~2.4 between 8,200 and $4.9 \times 10^4\,M_\odot$. Those ratios are independent of $m_{\rm eff}$, so the pre-registered deliverable below is unaffected by the correction; only the detectability margin moves. Resolving it requires measuring the reflex motion of the innermost stellar distribution at the few-μas yr^{-1} level against the bulk cluster frame: the regime of Roman's wide-field astrometry, which delivers ~10 μas-class differential astrometry over fields vastly larger than HST's, tying the inner arcseconds to thousands of cluster reference stars at once (WFIRST Astrometry Working Group et al., 2019). Two saturation caveats bound the reference-star pool: ω Cen's brightest giants saturate Roman's detectors even in the shortest exposure modes, and the faint end is photon-starved in crowding, so we take the usable reference-star window to be roughly F146 ≈ 16–21 mag (a stated assumption to be verified against the as-flown detector performance; the $\gtrsim 10^3$ stars-per-field figure below already assumes it). With launch announced for 2026 August 30, a five-year cadenced guest-observer program (ω Cen lies outside the core community surveys, so epochs must be requested, but the per-epoch cost is small) delivers, as its headline result, the factor-2.4 discrimination between the light-IMBH and heavy-IMBH wander regimes, independently of individual-star orbits. In the measurement-noise-limited case the program would do better: distinguishing the 17 per cent light-IMBH amplitude difference requires the wander amplitude itself measured to $\lesssim 6$ per cent, which is a reference-frame rather than photon-noise problem, and with $\gtrsim 10^3$ cluster reference stars per field tying the inner-arcsecond frame at each of $\gtrsim 10$ epochs the per-epoch frame error averages well below the expected few-μas yr^{-1} wander signal, in principle supporting 3–5σ discrimination within the light-IMBH regime. We do not headline that figure, because the discrimination is limited by the N-body modelling of the stellar background. Before the pre-registration locks, that systematic will be converted from unquantified to measured: the wander prediction will be pushed through the González Prieto et al. (2025) N-body realizations and the model-to-model scatter quoted as the systematic floor. The factor-2.4 light-versus-heavy separation is what survives a several-fold error inflation, and it is the deliverable we pre-register, gated in Table 5 on Roman's as-flown crowded-field astrometric performance (which has never been measured on a field like this) alongside the F146 reference-window assumption above.

The astrometric channel itself now carries a proof of capability: Whitaker et al. (2026) used the same HST astrometric archive (2002–2023), extended with JWST imaging, to detect the reflex wobble of a main-sequence star orbiting an unseen $4.46^{+1.22}_{-1.01}\,M_\odot$ companion, the first stellar-mass black hole confirmed in ω Cen and the first astrometric black-hole detection in any globular cluster. A dynamically measured dark remnant in the same field, with a 94-yr period and 31-AU separation recovered from curvature over a two-decade baseline, demonstrates end-to-end that the catalogue's astrometry supports orbit-grade inference at the precision this program assumes.

(ii) The absolute reference frame (Gaia DR4, 2026 December). The dominant systematic in all current inner-field astrometry is the tie between HST's relative frame and the absolute (ICRS) frame. Gaia DR4's 5.5-year solutions for the $r > 10''$ cluster members (Gaia Collaboration et al., 2016) improve proper motions by several-fold over DR2-era ties, propagating directly into the fast-star velocity vectors — and thus into the 8,200 $M_\odot$ lower bound itself, which rests on those velocities exceeding the escape speed. This is the cheapest decisive measurement in the entire campaign: a re-derivation of the Häberle et al. (2024b) bound on the DR4 frame either firms the tension into a $> 5\sigma$ contradiction with the timing bound or dissolves it. Deliverable in 2027 from archival data alone.

(iii) Inner-star discovery (ELT/MICADO, 2029+). MICADO's ~10 mas resolution and 39-m aperture (Davies et al., 2021) penetrate the central arcsecond at magnitudes HST cannot reach in crowding, with the realistic prize being new fast stars at $r < 0.02$ pc. A single star at 0.01 pc raises the acceleration signal of Eq. (2) by a factor of 64, converting Figure 2's verdict from "decades" to "a few epochs" — this is how the Galactic-Centre program succeeded (GRAVITY Collaboration et al., 2018), and ω Cen is its natural successor target at $10^2\times$ lower mass. With ELT first light scheduled for 2029 and commissioning occupying the first year of operations, the first MICADO astrometric epochs realistically arrive in 2030–31, at two epochs per year thereafter; first acceleration-grade results by the early 2030s.

5.3 Decision criteria

DR4 re-derivation (2027): if the fast-star velocity excess survives at $> 5\sigma$, H_1 requires the MSP analysis to be systematically biased — a specific, checkable claim (radial distribution priors; Bañares-Hernández et al. 2025). Roman wander (2031): light/heavy discrimination feeds the LISA prior. MICADO inner stars (2030s): any star with measured Keplerian curvature yields $M_\bullet$ to tens of per cent, independent of statistical modelling, before LISA flies.

6 Program 4: The LISA forecast

Bottom line. The decisive observable is an extreme- or intermediate-mass-ratio inspiral caught in band, which would measure the central object's mass and spin regardless of its electromagnetic state. There is no current limit to state; the program is a forecast. The target precision is fractional mass to 10^{-3}–10^{-4} and spin to 10^{-3}, against a target-specific occurrence probability of $\sim 3 \times 10^{-7}$ over a four-year mission at the González Prieto et al. (2025) capture rate. A Tier-3 measurement feeds decision point D4 and adjudicates the managed-system hypothesis in the same instant it settles the mass tension.

6.1 Science case

LISA is the campaign's final stage: the only instrument that can measure the central object's mass and spin to high precision regardless of its electromagnetic state (Colpi et al., 2024; Amaro-Seoane et al., 2017). For a compact object inspiralling into a 10^4-class IMBH, matched-filter parameter estimation delivers fractional precisions of order 10^{-3}–10^{-4} on mass and 10^{-3} on spin (Babak et al., 2017; Amaro-Seoane, 2018), a deliberately conservative floor: Babak et al. (2017) reports fractional mass errors as tight as 10^{-6} for favourable configurations, and we quote the weaker end of their range throughout. At 5.49 kpc (versus the Gpc distances over which LISA expects to detect its EMRI population), any ω Cen inspiral in band during the mission would be loud, with matched-filter SNR of order 10^4–10^5 depending on companion mass (Section 6), making occurrence probability, not sensitivity, the limiting factor.

6.2 Occurrence rates

Event-rate estimates in the LISA literature imply IMRI formation rates per cluster of order 10^{-9}–10^{-6} yr^{-1} depending on IMBH mass, binary fraction, and core density (Mandel et al., 2008; Amaro-Seoane, 2018; Fragione et al., 2018; Arca Sedda et al., 2021). A target-specific value is now available and is three decades narrower: the ω Cen models of González Prieto et al. (2025), in which the IMBH grows chiefly by capturing 30–40 $M_\odot$ black holes (mean 31 $M_\odot$), give a present-day capture rate of $(4–8) \times 10^{-8}$ yr^{-1}. We adopt that figure, and the 31 $M_\odot$ companion it describes, where a single number is needed, and retain the generic range and its lighter 1.4 $M_\odot$ companion case as the class-level comparison; the two populations are not interchangeable, and pairing one rate with the other's waveform misstates occurrence by more than an order of magnitude, as the Tier-2 calculation below shows explicitly. Over a 4–10-year mission the generic range yields an in-band inspiral probability for ω Cen of $\lesssim 10^{-5}$, and the target-specific rate gives $\sim 3 \times 10^{-7}$ over four years: possibly enhanced by the cluster's unusually dense remnant population, but not by the four to five orders of magnitude required to make it likely.

The forecast is stated in three tiers. **Tier 1 (probable, no ω Cen source):** the deliverable is a resolved-source count, not a stochastic-foreground limit. ω Cen at 5.49 kpc, localized to a $\sim$ 10-arcmin core, places any in-band binary of the remnant population among LISA's individually resolved sources rather than folded into an unresolved galactic-confusion background; the deliverable this tier owes is a predicted count under H_0 and H_1 against the observed null, a stronger and more localized test of the extended-component density profile than a foreground amplitude would be. **Tier 2 (early-inspiral capture):** a compact object caught in the early, slowly evolving portion of an inspiral, in band for years to millennia before plunge depending on companion mass. This is not a distinct source class from Tier 3; it is the same inspiral intercepted earlier, and its detectability and occurrence pull in opposite directions, and both depend on which population is doing the inspiraling. For the target-specific 31 $M_\odot$ companion the González Prieto et al. (2025) rate is licensed for: $\mathcal{M}_c = 411.9\, M_\odot$, giving $\dot{f} = (96/5)\, \pi^{8/3} (G\mathcal{M}_c/c^3)^{5/3} f^{11/3} \approx 1.65 \times 10^{-12}$ Hz s^{-1} at $f_{\rm GW} = 2$ mHz, residence $f/\dot{f} \approx 38$ yr, and a target-specific Tier-2 occupancy $(4–8) \times 10^{-8}$ yr$^{-1} \times 38$ yr $\approx 2 \times 10^{-6}$; at 0.8 mHz the residence lengthens to ≈ 440 yr and the occupancy recovers to $\approx 3 \times 10^{-5}$. Detectability is overwhelming at either frequency, SNR of order 10^6 over a 4-yr mission against LISA's instrument-only $\sqrt{S_n} \approx 3 \times 10^{-20}$ Hz$^{-1/2}$ at 2 mHz. For the lighter 1.4 $M_\odot$ companion the generic class-level rate applies, not the target-specific one: $\mathcal{M}_c = 64.3\, M_\odot$, $h \approx 3.5 \times 10^{-19}$ at 2 mHz (orbital radius ≈ 0.03 AU, $\sim 10^2\, r_g$) and 5.49 kpc, residence ≈ 850 yr ($\approx 10^4$ yr at 0.8 mHz), SNR of order 10^5 over 4 yr against the instrument floor (even a 0.1 $M_\odot$ companion accumulates $\sim 7 \times 10^3$), and an occupancy $(10^{-9}–10^{-6}\, \text{yr}^{-1}) \times (10^3–10^4\, \text{yr}) \approx 10^{-6}–10^{-2}$, derived from, not independent of, the Tier-3 rate. That 1.4 $M_\odot$ range is the generic-rate case, not one González Prieto et al. (2025) licenses a target-specific rate for; quote the band with both the number and the companion mass. The instrument-only $\sqrt{S_n} \approx 3 \times 10^{-20}$ Hz$^{-1/2}$ used above omits the galactic-confusion foreground, which for a 4-yr mission (Robson et al., 2019) adds 78 per cent in power at 2 mHz ($\sqrt{S_n} \approx 4.3 \times 10^{-20}$, SNR still of order 10^5 for the 1.4 $M_\odot$ case) and dominates by a factor 3.4 at 0.8 mHz ($\sqrt{S_n} \approx 4.3 \times 10^{-19}$), so the low-frequency end of the occupancy range above is bought at a sensitivity cost the raw residence numbers do not price. If any compact object is in band LISA cannot miss it, and the orbital frequency and its drift then measure $M_\bullet$ to precision intermediate between timing and a full late inspiral; but the expectation remains that the band is empty. **Tier 3 (lucky):** a genuine inspiral; per-mille mass and spin. Even in Tier 1, a spin constraint for the class arrives statistically: LISA's detected population of comparable IMBH systems at cosmological distances calibrates the natural spin distribution against which any eventual ω Cen measurement is judged. The campaign's structure ensures that even Tier 1 is decisive when combined with the pulsar acceleration map (Section 4): a resolved-source non-detection plus a point-mass timing profile isolates H_0. **Tier 4 (H_1's own mHz signature):** a fourth outcome belongs on this tier list even though it never activates H_2: a remnant subcluster dense enough to explain H_1 contains a substantial stellar-mass black-hole population, and that population produces its own resolvable mHz binaries (SNR $\sim$ few$\times 10^4$ at 2 mHz over a 4-yr mission), a distinct and unambiguous LISA signature that a null central massive object does not otherwise predict.

6.3 The spin measurement as the H_2 adjudicator

Paper A dependence (labelled): under H_2 the single sharpest residue is near-extremal spin ($a_\star \gtrsim 0.9$) on a hole that has demonstrably accreted nothing for Gyr; under H_0 natural formation channels for cluster IMBHs span low-to-moderate spin, and for ω Cen specifically the growth history modelled by González Prieto et al. (2025) ($\sim 10^3$ captures of 30–40 $M_\odot$ black holes onto a 500–5000 $M_\odot$ seed, seed-epoch mass ratio $q \simeq 0.006$–0.08, final-mass ratio $\mu/M_f \simeq 7 \times 10^{-4}$) is the isotropic minor-merger regime, which under isotropic capture ($\langle \cos\iota \rangle = 0$ for the captured objects) random-walks the spin to $a_\star \sim 0.05$–0.10 (central ≈ 0.06; Swanson 2026a) rather than toward the $\chi \simeq 0.7$ attractor of comparable-mass hierarchical growth, which the random walk alone reaches only at $q \approx 0.12$, a factor ~ 175 above ω Cen's final-mass ratio. That prediction holds only while the net-alignment fraction of the captured population stays below $\epsilon_{\rm rot} \approx 0.035$; above it a coherent, unsuppressed spin-up term takes over and the low-spin prediction degrades, a premise Swanson (2026a) states explicitly and recommends checking directly against the González Prieto et al. (2025) realizations. The contrast this test resolves is therefore the full low-versus-high one at this target, sharper than an earlier draft's $\simeq 0.7$-versus-$\gtrsim 0.9$ reading; the attractor caveat applies to the class-level LISA population, not to ω Cen. A Tier 3 spin measurement therefore adjudicates H_2 at the same instant it completes the H_0/H_1 question, the entire reason this campaign's electromagnetic and dynamical groundwork must precede LISA rather than follow it. The full decision tree, including this branch and its isotropy premise, is assembled in Section 13.

6.4 Deliverables before launch

Between now and the mid-2030s the program's work is preparatory but concrete: (i) maintain the dynamical model of the inner parsec (timing + astrometry) so that any LISA source has an immediate host context; (ii) publish the ω Cen-specific EMRI/IMRI rate forecast with the post-2026 mass constraints folded in; (iii) ensure the cluster's barycentric ephemeris and distance (± 0.06 kpc; Häberle et al. 2025) are maintained at the precision LISA parameter estimation will assume.

7 Program 5: KM3NeT/ARCA neutrino monitoring

Bottom line. The decisive observable is a high-energy neutrino multiplet: at least three tracks within 1° of the cluster, $E \gtrsim 10$ TeV, inside one sliding window, significant beyond 5σ post-trials with no astrophysical counterpart. The current best is the archival ANTARES+IceCube baseline, with ARCA at 51 of 230 detection units and sensitivity growing roughly linearly through the decade. The target is full-array steady-state point-source sensitivity of a few $\times 10^{-12}$–10^{-11} TeV cm^{-2} s^{-1} plus a decade-scale burst pipeline over $\sim 3 \times 10^6$ windows. A null closes the burst-mode channel at its stated strength; any candidate triggers the two-messenger protocol of Section 11.

7.1 Science case and geometry

ω Cen at declination -47.5° is an up-going source for the Mediterranean: KM3NeT/ARCA observes it through the Earth, with atmospheric muons filtered out and sub-0.2° angular resolution for track events at $\gtrsim$10 TeV (Adrián-Martínez et al., 2016). For IceCube the same source is down-going (degraded but usable above $\sim$100 TeV; Aartsen et al. 2020); the combined ANTARES+IceCube southern-sky analysis (Albert et al., 2020) defines the archival baseline. ARCA stood at 51 of 230 detection units as of early 2026, with real-time alert distribution entering commissioning; sensitivity grows roughly linearly with instrumented volume through the decade. The partially built array has already delivered a discovery-class

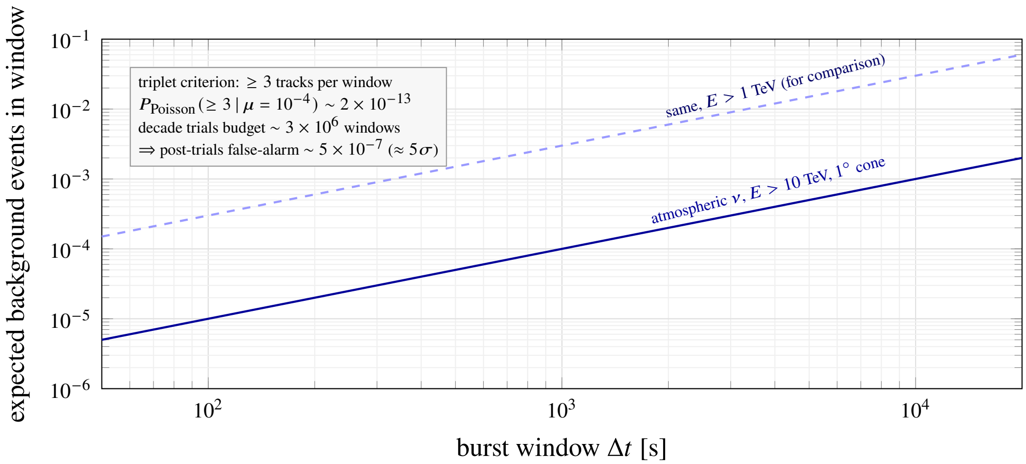

Figure 3: Design of the neutrino burst search. Expected atmospheric-neutrino background within 1° of ω Cen versus window length, for the working $E > 10$ TeV threshold (solid) and a softer 1 TeV threshold (dashed). At 10^3 s the expectation is $\sim 10^{-4}$ events, so the pre-registered triplet criterion is intrinsically secure against background even after a decade-scale trials budget; rate normalizations are order-of-magnitude, with exact values set by the detector Monte Carlo at analysis time.

result: the ~220-PeV track event KM3-230213A, the most energetic neutrino yet observed (The KM3NeT Collaboration, 2025), demonstrates that ARCA detects and reconstructs the rare high-energy events on which this program's burst criterion rests. The conventional science driver is a deep point-source limit on a dense old stellar system; the technosignature driver (Paper A dependence, labelled) is the burst-mode prediction of black-hole-based computation models (Dvali and Osmanov, 2023; Lacki and DiKerby, 2025), for which a cluster like ω Cen is the natural test bed and which the existing point-source pipelines do not target: their time-integrated statistics dilute rare short multiplets.

7.2 Implementation

Steady-state search (archival + ongoing). Standard unbinned-likelihood point-source analysis (Braun et al., 2008) at the ω Cen coordinates in (i) the ANTARES+IceCube combined dataset, (ii) the growing ARCA exposure. Full-array ARCA reaches $E^2\Phi \sim$ few $\times 10^{-12}$–10^{-11} TeV cm^{-2} s^{-1} at this declination over multi-year integrations — the deepest neutrino limit ever placed on a globular cluster.

Burst pipeline (new, the campaign's contribution). A pre-registered transient search in sliding windows of 10^2, 10^3, and 10^4 s within 1° of the cluster centre, $E \gtrsim 10$ TeV, on ARCA data plus IceCube alerts. The detection criterion (≥ 3 tracks in one window, $\geq 5\sigma$ post-trials, no plausible astrophysical counterpart) is calibrated by the background expectation: the atmospheric-neutrino rate above 10 TeV within a 1° cone is taken as 10^{-7} s^{-1} (~3 events yr^{-1} in the cone), a deliberate ~20× envelope on the ~0.15 yr^{-1} that Paper A derives for the same cone from the KM3NeT 2.0 letter of intent (Adrián-Martínez et al., 2016), as inherited via Swanson 2026b, and one that likely overstates the measured up-going rate for a cone this size by one to two orders of magnitude; even so, a 10^3-s window expects only $\sim 10^{-4}$ events and a triplet is intrinsically far beyond 5σ *before* trials; the design problem is the trials budget over a decade of monitoring ($\sim 3 \times 10^6$ windows), which the post-trials threshold absorbs (Figure 3). Any KM3NeT or IceCube real-time alert within 1° triggers the electromagnetic protocol of Section 11 — the alert system's first ω Cen-relevant cycle begins in late 2026.

The ≥ 3-track multiplet criterion reaches $p_{\text{det}} \approx 0.9$ only for burst fluences exceeding an RX-J1713-

equivalent steady source by $\sim 10^5$ (the ARCA effective-area integral, calibrated on the LOI's own event-rate table for the RX J1713 track analysis: 8.1 ν_μ tracks in 5 years after final cuts at a source declination of $-39.8°$, the same near-horizon visibility class as ω Cen at $-47.5°$). At astrophysically scaled fluences $p_{\rm det} \ll 1$, and the excluded burst rate scales as $1/p_{\rm det}$. The channel is therefore a test of an extraordinary hypothesis — burst fluences that no known astrophysical source produces — rather than a monitor of conventional neutrino emission.

7.3 Decision criteria

A decade-scale null at full-array sensitivity closes the burst-mode channel of Dvali and Osmanov (2023) for this system at its predicted strength — a publishable falsification regardless of H_2 (the underlying micro-black-hole physics is itself contested; Álvarez-Domínguez et al. 2024; Loeb 2024). One confirmed multiplet with no counterpart, repeated at the same location, would be among the most significant anomalies in high-energy astrophysics independent of any ETI interpretation; the protocol deliberately treats it as an astrophysical discovery first.

8 Program 6: archival Fermi-LAT and CTAO-South gamma rays

Bottom line. The decisive observable is a Fermi-LAT transient coincident with a Program-5 neutrino window, with a spectral excess above the few-GeV MSP cutoff and a CTAO TeV detection or limit as the conventional-science counterpart. The current best is the H.E.S.S.-era cluster survey limit. The targets are a sliding-window transient search over the full >17-year LAT exposure, any emission above ~30 GeV flagged as non-MSP, and a 50-h CTAO-South exposure reaching $\sim 10^{-13}$ erg cm^{-2} s^{-1} at 1 TeV. The program feeds the confirmation tier of the coordination protocol and screens dark-matter-versus-exotica readings before anomaly language.

8.1 Science case

ω Cen is a GeV source: 4FGL J1326.7−4729 coincides with the core, and its emission is attributed to the MSP ensemble after deep searches found no individual pulsations (Dai et al., 2020, 2023). This is both foreground and opportunity: the steady MSP emission is a calibration source against which *transient* or *spectrally anomalous* excesses can be sought with high contrast. The technosignature channel shares the burst phenomenology of Program 5 (the Dvali and Osmanov 2023 spectrum is "democratic" across species, implying γ-ray counterparts to any neutrino burst); the conventional channel is a TeV detection or limit on the cluster, of independent interest for MSP magnetospheric and inverse-Compton physics.

8.2 Implementation

Fermi-LAT (archival, immediate). Re-analysis of the now >17-year LAT exposure at the cluster position: (i) refit of the steady source with current diffuse models; (ii) a sliding-window transient search ($\Delta t = 10^2$–10^4 s, matched to Program 5's windows) over the full mission, with the steady MSP flux as null model; (iii) spectral-cutoff analysis: MSP ensembles cut off at a few GeV, so any significant emission above ~30 GeV flags a non-MSP component. All three are pure archival analyses.

CTAO-South (2027+). The first Paranal telescopes are, on early-2026 construction schedules, arriving through late 2026 (Cherenkov Telescope Array Consortium et al., 2019); intermediate arrays operate while construction proceeds. We propose (i) a 50-h deep exposure reaching $\sim 10^{-13}$ erg cm^{-2} s^{-1} at 1 TeV (~ 0.2 per cent of the Crab Nebula flux at 1 TeV on the HEGRA Crab spectrum, Aharonian et al. 2000) for the completed Alpha-configuration array; the 2027-era partial array delivers a factor 2–4 less, and the depth quoted here is recovered as construction completes. Either figure substantially betters the archival VHE limits placed on globular clusters by the H.E.S.S. cluster survey (H.E.S.S. Collaboration et al., 2013), making this the first CTAO-depth TeV limit on any globular cluster core. We also propose (ii) a standing target-of-opportunity trigger: any Program-5 alert within 1° pre-authorizes prompt CTAO follow-up, the latency being minutes for a source that transits Paranal $\sim 23°$ from zenith.

8.3 Decision criteria

A LAT transient coincident with a neutrino window elevates any Program-5 candidate to the campaign's highest tier. A TeV detection of steady emission is conventional astrophysics: it would make ω Cen the second globular cluster with detected VHE emission, after H.E.S.S.'s detection coincident with Terzan 5 (Abramowski et al., 2011), and the first with a resolved connection to a candidate IMBH environment. Its absence at CTAO depth tightens the inverse-Compton budget of the MSP population well beyond the existing H.E.S.S.-era cluster limits (H.E.S.S. Collaboration et al., 2013). Spectral excess above 30 GeV without transient behaviour prompts a dark-matter-versus-exotica analysis before any H_2 language is entertained.

9 Program 7: optical/IR time domain with Rubin/LSST and archival anomalies

Bottom line. The decisive observable is a nuclear tidal-disruption event, the one natural signal that would announce the central object unambiguously, together with a population-level photometric anomaly class at $\geq 3\sigma$ from the archival census. There is no current detection; the oMEGACat archive is essentially unmined for variability. The targets are an LSST broker filter flagging nuclear transients at $r \gtrsim 16$ and all-sky coverage of the saturated bright regime, plus a variability census of $\sim 10^6$ stars to ~ 0.05 mag depth over two decades of baseline. A TDE feeds the pre-arranged multi-wavelength response; anomaly classes feed the coordination protocol.

9.1 Science case

Two distinct searches share this program. First, transients: tidal disruption of a star by an IMBH is the one natural event that would announce the central object unambiguously (Hills, 1988; Gezari, 2021); rates are low ($\sim 10^{-7}$–10^{-6} yr^{-1} per cluster for full disruptions, higher for partial-disruption flares of the dense remnant environment) but the cost of monitoring is now effectively zero, because Rubin has taken science data since late 2025 (Pre-LSST), the ten-year LSST formally began on 2026 June 30, and ω Cen sits in the survey footprint (Ivezić et al., 2019). Second, archival photometric anomalies: the oMEGACat HST archive (500+ epochs over 20+ years for 10^6 stars; Häberle et al. 2024a; Nitschai et al. 2023) is an essentially unmined time-domain dataset in which periodic dimmings, secular fades, or statistically anomalous variability classes (the photometric residues conjectured for megastructures; Wright et al. 2014; Hsiao et al. 2021; Socas-Navarro et al. 2021) can be sought with two decades of baseline for free.

9.2 Implementation

LSST stream and bright-transient coverage. At the cluster's distance modulus of $\mu = 13.7$, a full TDE peaking at $M_V \approx -14$ to -17 would appear at $V \approx 0$ to -3: hopelessly saturated for LSST difference imaging, and indeed for any survey-class imager. The bright-event trigger therefore belongs to

ASAS-SN/Evryscope-class all-sky photometry, which monitors this magnitude range continuously; the campaign's role is to register the ω Cen position with those pipelines. Both instruments saturate well short of the TDE peak: ASAS-SN saturates at $V \approx 10$–11 mag (Kochanek et al., 2017), and Evryscope's short-exposure photometry is saturation-limited near $V \approx 6$ mag (Law et al., 2015), so the bright-transient trigger relies on the recovered, non-linear photometry of saturated frames rather than on standard aperture photometry. LSST covers the complementary faint regime: a standing broker filter on the field flags (i) any nuclear transient in the unsaturated range $r \gtrsim 16$, which captures partial disruptions and other sub-luminous accretion flares, and (ii) anomalous variables among $\sim 10^5$ measurable members outside the saturated core. Difference-imaging in the crowded annulus is handled with the oMEGACat reference catalogue.

Archival mining. A systematic variability census of the oMEGACat photometric archive: period search to ~0.05 mag depth across 10^6 stars, secular-trend extraction, and an explicitly pre-registered anomaly statistic (population-level excess of non-physical light-curve classes) with human vetting of survivors. Conventional yield: the deepest variable-star census of any globular cluster, eclipsing-binary distance cross-checks, and stellar-rotation demographics — publishable regardless of anomalies.

Core-depletion statistic (joint with Program 3). Paper A dependence (labelled): H_2's P4c predicts secular depletion of the loosely bound core population beyond N-body expectations. The measurement (star counts by binding-energy quantile against matched N-body models; González Prieto et al. 2025) is identical to standard mass-segregation analysis and is conventional dynamics on its face; it operationalizes Paper A's P4c observable, stated there as a mass-function-slope test, as a binding-energy-quantile star count instead. We do not pre-register the depletion direction as a primary discriminant: at the 10-pc delivery radius the Phase-4 deficit is ~1 per cent of the enclosed mass, the removed mass reappears as central point mass, and any deficit refills on the ~4-Gyr core relaxation time (Swanson, 2026a).

9.3 Decision criteria

A nuclear TDE settles the IMBH question electromagnetically (and its decay light curve constrains $M_\bullet$; Gezari 2021); the campaign's role is to guarantee the multi-wavelength response is pre-arranged. Archival anomalies are population statistics: no single weird star means anything (the false-positive history of photometric SETI is long); only a $\geq 3\sigma$ class-level excess, surviving vetting, enters the coordination protocol.

10 Program 8: Supplementary archival channels

Bottom line. The decisive observable is an X-ray point source at the kinematic centre, at any flux; continuous-wave searches at twice the known pulsar spin frequencies and a stacked ultra-high-energy cosmic-ray directional analysis complete the messenger coverage. The current best is the ~291 ks Chandra limit of $L_X \lesssim 1.6 \times 10^{30}$ erg s^{-1}. The targets are a re-derivation of that limit at the current centre, stacked Chandra/XMM-Newton depth, an eROSITA variability screen, O5-class CW limits of $h_0 \lesssim 10^{-26}$, and a public-data UHECR stack. The X-ray result feeds the joint fundamental-plane constraint with Programs 1 and 2.

Three channels are cheap enough to run as graduate-student archival projects and complete the messenger coverage.

Archival X-rays (Chandra, XMM-Newton, eROSITA). The X-ray band is the standard probe of quiescent black-hole accretion, and the campaign as presented so far would otherwise lack it. The benchmark limit is the ~291 ks Chandra exposure of Haggard et al. (2013), which detects no central point source and bounds the unabsorbed luminosity at $L_X(0.5\text{–}7.0\ \mathrm{keV}) \lesssim 1.6 \times 10^{30}\ \mathrm{erg\ s^{-1}}$, an Eddington ratio of $\lesssim 10^{-12}$ for a $10^4\ M_\odot$ hole. The archival program re-derives this limit at the current oMEGACat kinematic centre (the 2013 analysis predates the fast-star centre determination), stacks the subsequent Chandra and XMM-Newton exposures of the field, and folds in the released eROSITA all-sky survey epochs (eRASS1 in the DR1 release, plus the subsequent completed passes as the consortium releases them) as a variability screen, noting that the survey was halted in February 2022, so the eROSITA screen covers 2020–2022 only; a joint X-ray–radio treatment through the fundamental plane (Merloni et al., 2003) then constrains radiatively inefficient flow models alongside the Program 1 and Program 2 limits. Any X-ray point source at the kinematic centre, at any flux, would be a major result; the expected null sharpens the deepest multi-band quiescence measurement for any black hole candidate.

Continuous gravitational waves (LIGO–Virgo–KAGRA). Narrowband searches at twice the spin frequencies ($f_{\mathrm{GW}} = 2f_{\mathrm{spin}}$) of the nineteen known MSPs, using public O5 data and the MeerKAT ephemerides, following the known-pulsar methodology of Abbott et al. (2022). Expected upper limits $h_0 \lesssim 10^{-26}$ are an O5-class expectation (a factor $\gtrsim 2$ better than the best O3 targeted result, PSR J0711−6830, once converted to this frequency band) and translate, at ω Cen's own MSP population, which clusters near $P \approx 4$ ms ($f_{\mathrm{GW}} \approx 500$ Hz), to ellipticity limits $\varepsilon \approx 2.1 \times 10^{-7}$ at 5.49 kpc. Reaching a few $\times 10^{-8}$ needs a faster spinner than any known ω Cen MSP: at $P = 2.5$ ms ($f_{\mathrm{GW}} = 800$ Hz) the limit is $\varepsilon \approx 8 \times 10^{-8}$, and 3×10^{-8} requires $P \approx 1.5$ ms. This is conventional neutron-star physics in an environment (a dense old cluster) where the recycled-pulsar population is dynamically distinctive, with one complication the standard pipeline does not carry: the observed $\dot{f}$ of a cluster MSP is dominated by line-of-sight acceleration in the cluster potential rather than intrinsic spin-down, several ω Cen MSPs have measured negative $\dot{f}$ for exactly this reason, which Program 2's own timing-acceleration map exploits as signal rather than noise. The search must therefore fit $\dot{f}$ and $\ddot{f}$ as free terms rather than assume a spin-down model, and the spin-down-limit framing standard targeted searches quote is not available for these targets. The trials factor across the nineteen targets is carried in the post-trials threshold of Table 5. The technosignature reading (structured masses in MSP orbits) is not separately funded; it shares the identical data product.

Ultra-high-energy cosmic rays (Pierre Auger). ω Cen sits well inside Auger's southern acceptance. A stacked directional analysis around the cluster position in the public $E > 8$ EeV dataset (Pierre Auger Collaboration et al., 2017), with 1° and 3° apertures bracketing magnetic deflection, either contributes an upper limit on hadronic acceleration in the core (of conventional interest given the MSP wind population) or flags an anisotropy for which the Galactic-Centre region (52.4° away, but the dominant anisotropy structure at these energies) is the confounder to exclude first.

11 Coordination, joint statistics, and data policy

11.1 The two-messenger rule, operationalized

Table 5 pre-registers the campaign's trigger and confirmation matrix. Each row names a primary trigger, the channels that can confirm it, and the joint-significance requirement. Combined significances across independent channels use Fisher's method on the per-channel p-values; each per-channel p-value entering the combination must itself be post-trials within its channel (its look-elsewhere budget over windows, frequencies, and positions already paid), so the combination never launders pre-trials significances. Two opposing effects enter the combined significance; both are stated. Because post-trials p-values are conservative rather than uniform under the null, the Fisher combination inherits that conservatism. But Fisher's method assumes the inputs are independent under the null, and several of the pairings in Table 5 are not: a genuine tidal-disruption flare excites P1, P2, P7 and P8 together, and the LAT and CTAO rows share both the Galactic diffuse model and the cluster's own millisecond-pulsar population. Under positive dependence the Fisher statistic has larger variance than the χ^2_{2n} reference, so the tail probability is understated and the method is anti-conservative. The two effects have no reason to cancel. We therefore apply Brown (1975) for any combination whose inputs are not demonstrably independent under the astrophysical null, estimating the null covariance of the per-channel statistics from archival data (which costs no observing time) and rescaling the degrees of freedom accordingly; the covariance-estimation procedure is pre-registered alongside the thresholds, though the specific archival datasets it will draw on are not yet selected and this element of the plan remains aspirational. Where the covariance is not yet estimated, the combination is restricted to channel pairs that are independent by construction, and the stated global threshold bounds rather than equals the nominal false-alarm rate. The 5×10^{-7} figure is a per-trigger-class threshold, not a campaign-wide false-alarm budget over the ten trigger rows and fifteen years of monitoring; a campaign-wide false-alarm budget aggregating all ten channels is not computed in this paper and is deferred to the individual pipeline papers. The global claim threshold is set at $p < 5 \times 10^{-7}$ post-trials — deliberately at discovery convention, because the base rate of true technosignatures is unknown and plausibly zero, so the loss function is asymmetric: a false positive damages the field's credibility far more than a delayed true positive damages the discovery (Wright et al., 2022).

This matrix is one of two pre-registrations governing the campaign, and the two occupy different layers. The Fisher-combined $p < 5 \times 10^{-7}$ matrix of Table 5 is the *trigger layer*: it decides when an event or excess is promoted from a single-channel candidate to a campaign-level anomaly requiring adjudication. Paper E's hierarchical Bayes-factor framework, with Kass–Raftery $\ln K$ action bands (Kass and Raftery, 1995) whose "candidate" band itself requires evidence in at least two messengers, is the *inference layer*: it takes any promoted trigger and adjudicates it against the full $H_q/H_{\text{sub}}/H_{\text{eng}}$ hypothesis set (Swanson, 2026a). The frequentist matrix therefore controls the false-alarm budget of what enters adjudication, while the Bayesian framework determines what the adjudicated evidence supports; a trigger that clears Table 5 but lands in a weak $\ln K$ band is reported as an unexplained astrophysical candidate, not an anomaly claim. One mandatory rule governs the interface between the layers. Promotion is data-dependent, so the inference layer always evaluates its likelihoods on the complete campaign record, every epoch and every channel including those that recorded nothing, never on the promoted subset alone. The trigger layer allocates follow-up effort and carries no inferential weight. Scoring only the promoted data would require an explicit selection correction of order $\ln P(\text{trigger} \mid H_0) \approx -14$ nats at the threshold above, which is large enough to move a nominal candidate to null-favoured; carrying the full record avoids the correction rather than paying it (Swanson, 2026a). The significance conventions of the two layers, and of the companion papers that consume or produce them, are set side by side in Appendix 14.

11.2 Alert latency and standing authorizations

The latency-critical path is neutrino → gamma/optical: KM3NeT real-time alerts (commissioning through 2026) distribute within seconds–minutes; the CTAO ToO and LSST-broker hooks are standing authorizations requiring no human decision for the first response hour. Slow-path coordination (JWST DDT requests, radio re-pointing) is pre-drafted as template proposals with trigger criteria attached, cutting submission latency to <24 h.

11.3 Data policy

All pipelines, thresholds, and analysis code are public from the start (the pre-registration is itself the methods paper); all derived catalogues are released with their papers; raw data inherit host-facility policies. That guarantee covers what the campaign controls: its own code, thresholds, and derived products. What

Table 5: Pre-registered trigger and confirmation matrix. “Joint” = Fisher-combined post-trials significance required for any anomaly claim; single-channel events, however significant, are reported as astrophysical candidates only.

Primary trigger	Channel threshold	Confirming channels	Disposition
Neutrino multiplet (P5)	$\geq$ 3 tracks / 1° / $10^{2–4}$ s / $\geq$10 TeV, 5σ post-trials	Fermi-LAT window (P6); CTAO ToO (P6); JWST/ground IR (P1)	repeat at same position required for anomaly claim; single event published as astrophysical transient
Narrowband radio line (P2)	SNR > 15; multi-beam cross-rejection; multi-epoch scintillation-aware confirmation (Section 4)	off-source cadence; second telescope (e.g. ATCA)	interference exclusion protocol precedes any claim
IR variable/transient (P1, P7)	$\geq$ 20% amplitude or new source > 5σ	radio continuum (P2); X-ray archival; LSST stream (P7)	classified as accretion/TDE candidate first
Photometric anomaly class (P7)	$\geq 3\sigma$ population-level excess	independent archive (Gaia epochs); spectroscopy	population statistic only; no single-star claims
Astrometric anomaly (P3)	wander outside H_0/H_1 envelopes at 3σ; gated on verified Roman crowded-field astrometric performance and the F146 ≈ 16–21 reference window (core-depletion statistic retired as a primary discriminant: at the 10-pc delivery radius the Phase-4 deficit is ∼1 per cent of the enclosed mass, the removed mass reappears as central point mass, and any deficit refills on the ∼4-Gyr core relaxation time; Harris 1996; Swanson 2026a)	timing acceleration map (P2)	feeds LISA prior; no standalone claim
>30 GeV spectral excess (P6)	$\geq$ 5σ non-MSP component in LAT refit	CTAO follow-up (P6); radio/X-ray positional coincidence (P2, P8)	dark-matter-versus-exotica analysis precedes any anomaly language
CW detection at MSP frequency (P8)	$\geq 5\sigma$ in known-pulsar narrowband search	EM timing solution and position (P2)	classified as neutron-star physics first
UHECR anisotropy (P8)	$\geq$ 3σ post-trials stacked excess, 1°–3° apertures	Fermi-LAT/CTAO coincidence (P6); Galactic-Centre confounder excluded	directional statistic only; no source claim without γ-ray support
X-ray point source at centre (P8)	any source > 3σ at kinematic centre	radio continuum (P2); JWST IR counterpart (P1)	classified as accretion candidate; feeds fundamental-plane fit
LISA spin/mass result (P4)	per mission parameter estimation	all of the above as context	adjudicates $H_0/H_1/H_2$ per Section 13

Table 6: Campaign summary. "Type": A = archival/commensal, N = new observations, F = free by-product of funded survey. Costs are rough full-cost estimates (Section 12); facility time is listed separately since it is allocated, not purchased.

#	Program	Type	Key sensitivity	Facility request	Timeline	Decides
P1	JWST IR limits	N	~10 nJy (F444W); $L_{\rm bol} \lesssim 10^{30}$ erg s^{-1}	~65 h, 2 epochs	2027–2029	accretion state; waste heat
P2	Radio: timing + SETI + continuum	N	$\sigma_a \sim 10^{-10}$ m s^{-2} (5 yr); EIRP 4×10^{15} W (incoh., full cluster)	~130 h yr^{-1} MeerKAT, 5 yr	2026–2031+ → SKA-Mid	H_0 vs H_1 (profile); first SETI limit
P3	Astrometry: DR4 + Roman + ELT	A/F/N	frame to DR4; wander factor-2.4 (light vs heavy); inner-star discovery	archival; Roman survey; ~28 h ELT	2026–2035	mass tension; wander mass
P4	LISA forecast	F	$\delta M/M \sim 10^{-3\text{–}4}$, $\delta a_\star \sim 10^{-3}$ (if inspiral)	none (mission science)	2035+	everything (Tier 3)
P5	Neutrino monitoring	A/N	$E^2\Phi \sim$ few$\times10^{-12}$ TeV cm^{-2} s^{-1}; burst triplets	pipeline on ARCA stream	2026–2040	Dvali–Osmanov channel
P6	Gamma rays	A/N	LAT transient windows; 10^{-13} erg cm^{-2} s^{-1} at 1 TeV (50 h)	50 h CTAO + ToO	2026–2032	burst counterparts; first CTAO-depth TeV limit
P7	Time domain	A/F	TDE flares unmissable; 10^6-star archival census	LSST stream; archival	2026–2036	TDE; anomaly classes
P8	X-ray + CW + UHECR archival	A	$L_X \lesssim 10^{30}$ erg s^{-1}; $h_0 \lesssim 10^{-26}$; stacked UHECR limit	public data	2026–2029	completeness

host collaborations must grant is a separate matter, negotiated by memorandum of understanding and costed in Table 7: MeerKAT and SKA raw data fall under SARAO proprietary policy, and KM3NeT does not release track-level data outside the collaboration, so the Program-5 burst pipeline runs either inside the collaboration on the full stream or externally on the much smaller public alert stream. Negative results are published on a fixed, pre-announced cadence, so that the decision to publish never depends on what the data showed.

11.4 Program summary

Table 6 assembles the eight programs with sensitivities, requests, and timelines.

12 Cost and timeline

12.1 Costing

Table 7 presents full-cost estimates (salaries, computing, travel, publication; US full-cost academic rates including overheads) for each program. Facility time is excluded (it is competitively allocated and carries no cash cost to the campaign), as is LISA, which is mission science. Two lines deserve comment. The data-management line prices the Program-2 commensal product, which is not free: the 1-Hz-resolution full-band spectrometer output is ~0.7 TB hr^{-1}, or ~90 TB yr^{-1} of reduced product at the requested cadence, plus transient voltage buffers, with archiving and transport budgeted at SARAO. The MoU line covers the host-collaboration data-access agreements of Section 11. With a 20 per cent contingency the total is ≈US$8.7M over a decade — still small by every relevant comparison: a single mid-scale NASA Explorer instrument, one JWST cycle's archival-funding pool, or the marginal cost of a few nights of ELT operations.

Table 7: Indicative full-cost budget (US$k, 2026 dollars). Personnel figures assume postdoc-led programs with PI/Co-I fractions and graduate students where noted. Estimates for the Roman component of P3 and for P5, P6, P7 are new to this paper; others follow the per-instrument proposal studies available at `https://omegacentauri.me`.

Program	Cost	Years	Dominant element
P1 JWST deep imaging + waste heat	1,310	3	postdoc + grad + HPC
P1b JWST spectroscopic phase (contingent)	257	2	postdoc (triggered only)
P2 MeerKAT timing + SETI + continuum	2,900	5	2 postdocs + grad + computing
P3 HST/Gaia frame + Roman wander	560	4	postdoc + HPC (full-cost rates)
P3b ELT/MICADO epochs	490	4	postdoc share + travel
P4 LISA preparatory modelling	150	3	PI/Co-I fractions
P5 Neutrino burst pipeline	380	3	postdoc + compute (full-cost rates)
P6 Fermi archival + CTAO ToO	200	3	postdoc share
P7 LSST broker + archival mining	300	3	postdoc + broker engineering
P8 X-ray + CW + UHECR archival	239	3	graduate RA + HPC
Data management (P2 commensal product)	450	5	~90 TB yr^{-1} archive + transport at SARAO
Host-collaboration MoUs (SARAO, KM3NeT)	120	10	data-access agreements
Coordination, data releases, workshops	150	10	part-time coordinator
Subtotal (P1b excluded)	7,249	—	—
Contingency (20%)	1,450	—	—
Total (P1b excluded)	**8,699**	—	—

12.2 Timeline

Figure 4 lays the programs against the facility schedule. The structure is deliberate: the 2026–2028 window is dominated by archival work and standing-pipeline construction (cheap, immediate); 2029–2034 by the new-facility harvest (ELT, SKA-Mid, CTAO completion, Roman mid-mission); 2035+ by LISA. Decision points D1–D4 are marked and defined in Section 13.

13 The decision structure through 2040

Figure 5 lays the facility schedule, the program phases, and the D1–D4 decision points on a single time axis; Figure 6 then assembles the campaign into a single decision tree, extending the gravitational-wave branch of Paper A's falsification framework backward into the electromagnetic-and-timing era. Rough branch weights are stated where the inputs exist to estimate them; they are planning priors, not forecasts, and we expect D1/D2 to revise them substantially. The tree's light-branch mass cutoff and spin boundary are taken verbatim from Paper A's falsification matrix (Table 3 of Swanson 2026b, tests T4 and T5); Paper A is the fence owner and no fence is restated independently here.

13.1 Amendment C-A1 (2026-08-26, pre-data)

An external audit round identified three defects in the decision tree as printed through Draft v2.3: the D4 mass partition left $[5 \times 10^3, 10^4)$ $M_\odot$ firing no branch at all, an interval that contains the fast-star kinematic lower bound of 8,200 $M_\odot$; the D2 routing into the extended, point-mass, and inconclusive branches carried only qualitative labels while the upstream mock-forecast gate was already numeric; and D1's binary split disclosed no basis for its 25/75 weights. All three are pre-registered-rule defects, so none is repaired

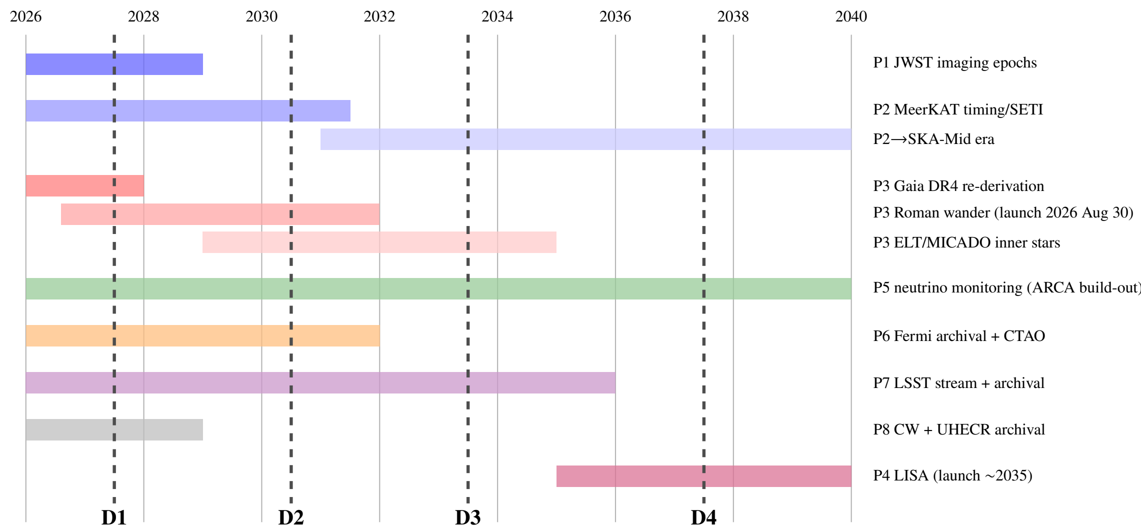

Figure 4: Campaign timeline against the facility schedule, 2026–2040. Decision points: D1 (2027): Gaia-DR4 re-derivation of the fast-star bound. D2 (2030): five-year pulsar acceleration profile; Roman wander first results. D3 (2033): pre-LISA synthesis — H_0 versus H_1 called at available significance. D4 (2037): LISA verdict, if a source is caught.

silently, following the disclosure convention of the mass-tension companion's Amendments A1–A5. No observation has yet been scored against the tree: every program in this paper remains pre-D1, and this amendment changes only the tree's own text, not any result.

The repair follows a ratified panel review (`board/notes/R7-PANEL-DECIDE.md`, brief B-1, option (a)) of three ledger findings (`C-M5`, `C-M6`, `C-t13`). First, the D4 gap is closed by splitting the light branch at Paper A's own T4 fence ($5 \times 10^3\ M_\odot$, adopted verbatim, not restated independently): masses below it retire H_2 as before, and the newly covered $[5 \times 10^3, 10^4)\ M_\odot$ interval is scored as a weakened, not falsified, selection ranking, exactly as Paper A's own T4 language already treats that mass range (a $5 \times 10^3\ M_\odot$ hole still delivers $\sim 6 \times 10^{34}$ W). Second, the D2 routing adopts the numeric thresholds $\Delta\chi^2 \geq +9$ (extended), $\Delta\chi^2 \leq -9$ (point mass), and $|\Delta\chi^2| < 9$ (inconclusive), taken from this paper's own pre-registered mock-forecast criterion (Section 4, `0xAlpha/results/R7-CALC-C2D.md`). That forecast has since been executed: at the real MSP sky positions and the current census of nineteen pulsars, the median simple-vs-simple discrimination between the two named potentials met the $\Delta\chi^2 \geq 9$ criterion with margin at every noise floor tested, from the timing-limited $10^{-10}\ \mathrm{m\,s^{-2}}$ floor to three times Paper H's spin-down-dominated budget; the equivalent evidence band is $\ln K \geq +4.5$ (extended) or ≤ -4.5 (point mass), since $\Delta\chi^2 = 2 \ln K$ for that forecast's fixed-parameter, simple-vs-simple statistic. The margin means the threshold is not chosen to make D2 easy to satisfy: it is met before any real data are scored, and the routing rule is adopted precisely because the forecast shows it discriminates. Third, D1's 25/75 split is left in place but is now stated explicitly as a disclosed planning prior with no claimed statistical derivation, matching this figure's own long-standing caption convention; the binary structure is retained because any excess that fails to survive the DR4 re-derivation at $> 5\sigma$ collapses to the same systematics-bias reading regardless of degree, not because a graded middle outcome was considered and discarded.

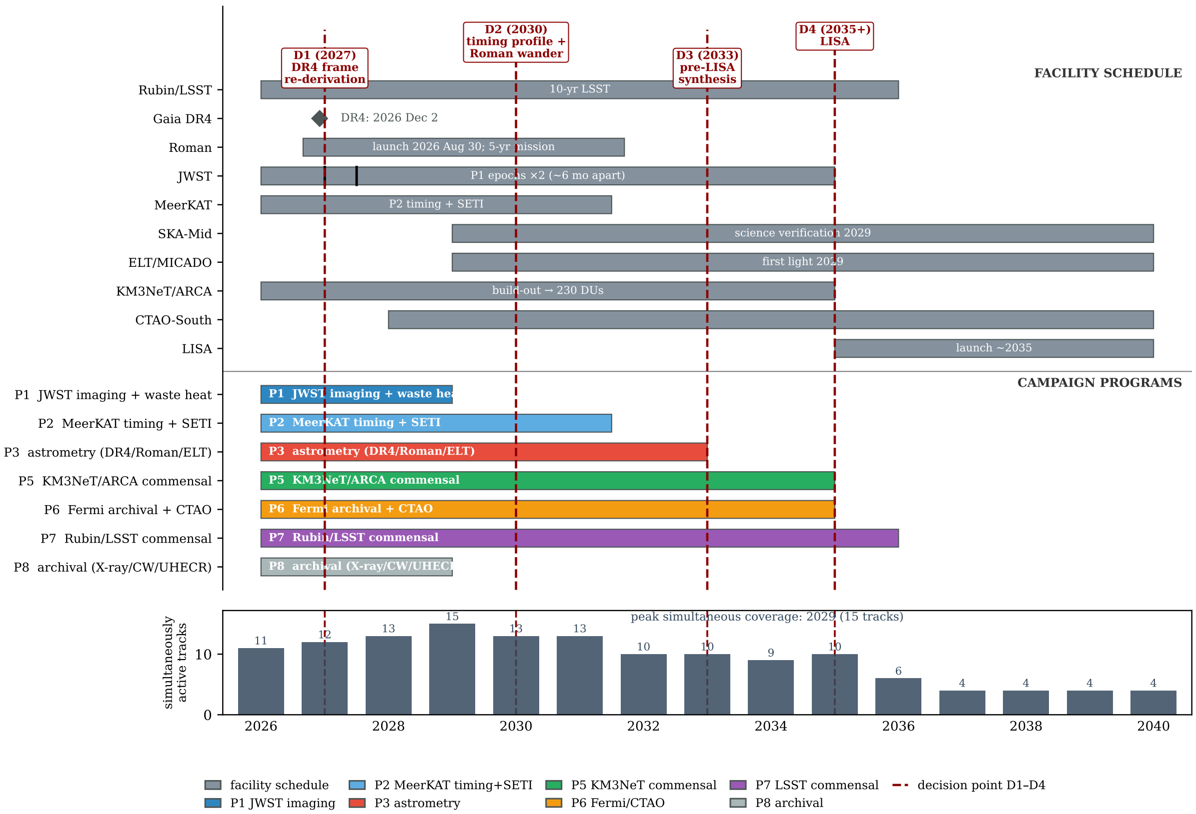

Figure 5: Campaign timeline as a Gantt chart, 2026–2040. Top block: the facility schedule (grey), with Gaia DR4 as a diamond marker at 2026 December 2, the two P1 JWST epochs as ticks on the JWST bar, the KM3NeT/ARCA bar ending at the 230-detection-unit build-out, and the LISA bar starting at the ~2035 launch. Middle block: the observing programs P1–P8 (Table 6), coloured as in Figure 4. Vertical dashed lines: the decision points D1 (2027, Gaia-DR4 frame re-derivation), D2 (2030, five-year timing profile and Roman wander), D3 (2033, pre-LISA synthesis), and D4 (2035+, LISA; Section 13). Bottom strip: the number of simultaneously active facility and program tracks per year. Coverage peaks in 2029, when SKA-Mid and ELT/MICADO come online while the MeerKAT, JWST, and KM3NeT programs are still running; the decisive measurements cluster at D1–D3 and at the LISA window.

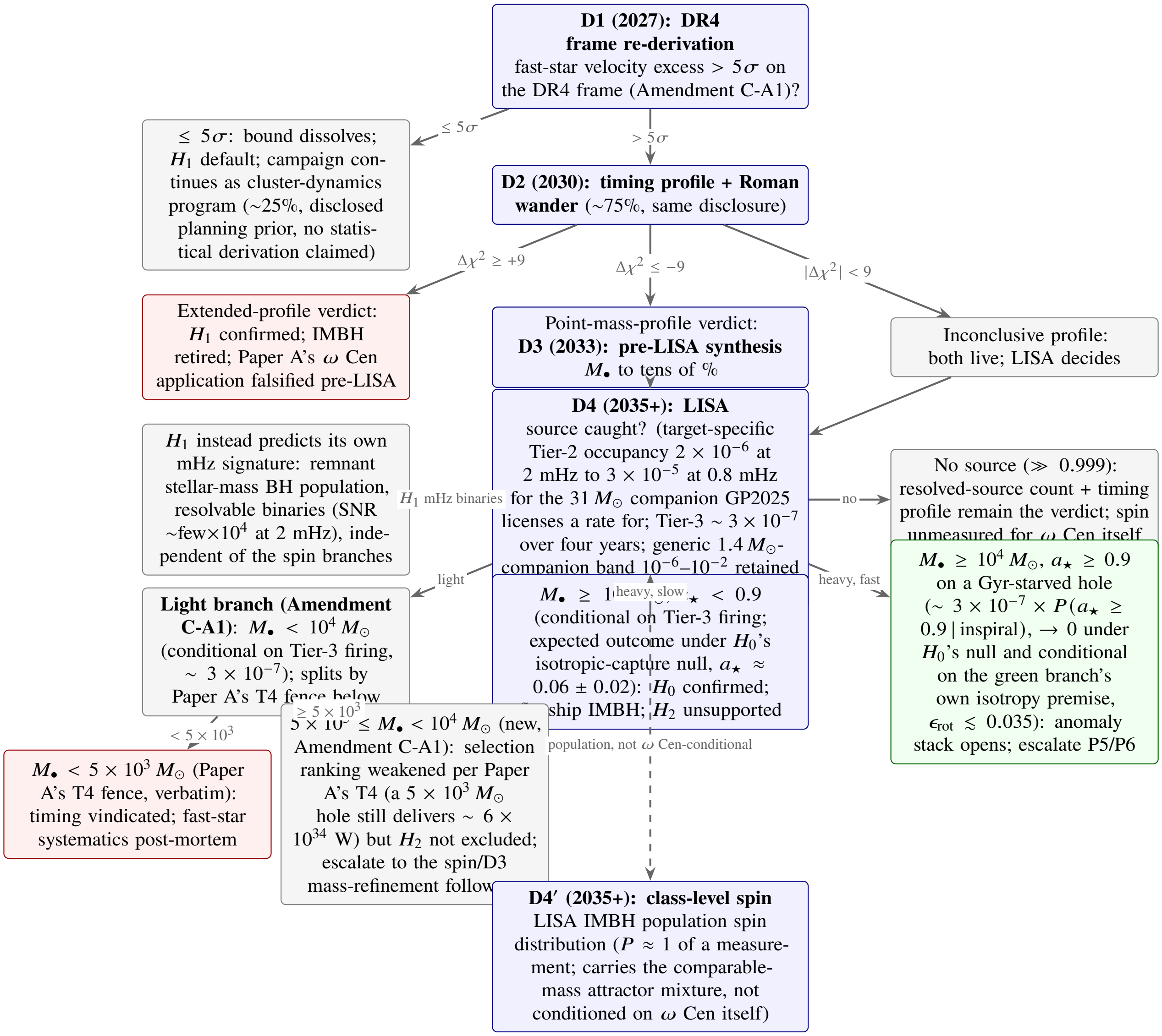

Figure 6: Decision structure through 2040. Red boxes retire hypotheses; the single green branch is the only path on which the technosignature interpretation gains material support, and even there the claim is "anomaly requiring explanation," conditional on the isotropic-capture premise stated on the branch. Percentages and weights are disclosed planning priors, not computed probabilities, stated at the target-specific rate where GP2025 licenses one; D1's split is binary because any excess at or below the 5σ contradiction threshold reverts to the same systematics-bias reading, not because a graded middle state was considered and discarded. D2's routing follows the numeric thresholds of Amendment C-A1 below, sourced to the executed mock forecast (`0xAlpha/results/R7-CALC-C2D.md`), whose statistic is signed positive toward the extended potential: $\Delta\chi^2 \geq +9$ (equivalently $\ln K \geq +4.5$ under that forecast's simple-vs-simple statistic, for which $\Delta\chi^2 = 2\ln K$) routes to the extended branch, $\Delta\chi^2 \leq -9$ ($\ln K \leq -4.5$) to the point-mass branch, and $|\Delta\chi^2| < 9$ to the inconclusive branch. D4's light branch now splits at Paper A's T4 fence ($5 \times 10^3\,M_\odot$), closing the previously uncovered $[5 \times 10^3, 10^4)\,M_\odot$ interval (Amendment C-A1); the new sub-floor node does not retire H_2, since Paper A's own T4 language already treats that mass range as weakened, not falsified. D4′ is the class-level spin test Paper E identifies as the campaign's most informative measurement; it is not ω Cen-conditional and is shown dashed for that reason. Every terminal node is a publishable scientific result about a first-rank astrophysical object.

14 Significance conventions across the companion papers

The campaign and its companion papers score evidence in four conventions, and a number quoted in one vocabulary must not be silently read in another. This appendix states the four side by side.

Post-trials p-values (this paper). The frequentist false-alarm probability of a trigger after the look-elsewhere correction over the search's windows, frequencies, and positions. The action boundary is $p < 5 \times 10^{-7}$ per trigger class (Table 5), the discovery convention, with cross-channel combinations handled as in Section 11. It applies to the trigger layer of all eight programs in this paper.

$\ln K$ **Bayes factors (Paper E).** The logarithm of the ratio of evidences between two hypotheses, taken by Paper E as $H_{\rm eng}$ against the best astrophysical null. The action bands follow Kass and Raftery (1995) as pre-registered in Paper E: below 0 null-favoured, 0 to 1 uninformative, 1 to 3 anomaly, 3 to 5 strong anomaly, and above 5, sustained across at least two messengers, candidate. The look-elsewhere treatment is structural: likelihoods are always evaluated on the complete campaign record, never on the promoted subset alone, so the selection effect is priced rather than corrected after the fact, and prior sensitivity is reported as a band rather than a point. It applies to Paper E's adjudication layer, acting on whatever this paper's trigger layer promotes.

95 per cent credible upper limits (Paper F). The Bayesian one-sided credible bound on a parameter, in Paper F the Bondi efficiency $\epsilon_{\rm B}$ or the dark mass. There is no action band; the limit is the product rather than a decision. The corresponding treatment is prior sensitivity, reported alongside the bound, since a credible limit conditions on the model instead of testing it. It applies to Paper F's quiescent-accretion limits on ω Cen and on the comparison clusters.

Posterior mass fractions (Paper H). The fraction of posterior probability in a region of the dark-component mass and scale-radius plane under one normalized model. There is no discovery threshold; the pre-registered criteria are consistency checks, sign stability of the Bayes factor across every prior cell and posterior-predictive adequacy in each cell. The prior-grid sweeps are themselves the sensitivity treatment, and a verdict is quotable only when it survives the full pre-registered cell set. It applies to Paper H's joint fast-star and pulsar fit.

No single convention is privileged; the choice depends on whether the question is "should we follow up?" (frequentist), "what does the evidence support?" (Bayes factor), "what is the bound?" (credible limit), or "is the model consistent?" (posterior check).

15 Conclusion

Omega Centauri in 2026 presents a configuration that observational astronomy rarely supplies: a contested discovery (the Galaxy's best IMBH candidate) whose two strongest constraints formally contradict each other; an electromagnetic silence deep enough to be a result in itself; an essentially untouched technosignature parameter space; and a fleet of new southern and all-sky instruments (Rubin, Roman, Gaia DR4, KM3NeT, CTAO, SKA-Mid, ELT, and ultimately LISA) arriving on the timescale needed to resolve all of it. The campaign presented here is designed to use that conjunction fully.

Its architecture reflects three commitments. First, *dual use*: every program stands as conventional astrophysics (the mass tension, the quiescent-accretion frontier, the MSP dynamical laboratory, the first deep TeV and neutrino limits on a globular cluster), so that no null result is a loss. Second, *conservative* *sensitivity accounting*: where the obvious measurement fails (direct fast-star acceleration before ~2040), we demonstrate the failure quantitatively and redesign around it, because a program that oversells its tests invites the credibility collapse that has repeatedly damaged this field. Third, *pre-registration*: hypotheses $H_0/H_1/H_2$, thresholds, confirmation rules, and decision points are fixed in advance, so that whatever the sky delivers (a remnant swarm, a quiet heavy hole, or a genuine anomaly), the inference chain was written down before anyone knew the answer.

The adjudication asymmetry deserves a blunt statement. Under realistic outcomes, this campaign will resolve H_0 versus H_1; it will neither support nor retire H_2 except through the low-probability LISA spin branch or a serendipitous multi-messenger multiplet. No null result carries evidential weight against H_2, because the leakage-free and electromagnetically silent predictions of that hypothesis class are compatible with every null the campaign can produce; what the campaign buys for H_2 is the pre-registered infrastructure to adjudicate a positive trigger if one ever arrives, and quantitative limits on the specific loud sub-classes (beacons, warm waste heat above the JWST floor, neutrino bursts at predicted strength).

The cost is US$8.7M across a decade including 20% contingency (Table 7), most of it salaries for archival analysis riding on facilities that are funded regardless. The payoff structure is asymmetric in the campaign's favour: the probable outcomes (a resolved mass tension, a characterized central object, the first technosignature limits on 10^7 old stars) are solid science at modest cost, and the improbable outcome (the green branch of Figure 6) would justify the program many thousands of times over. We commit, symmetrically, to the red branches: if the timing profile confirms an extended remnant component, or LISA delivers a light or slowly spinning hole, the anomaly hypotheses retire here without special pleading, and what remains is what was always underneath: the most interesting stellar system in the southern sky, finally measured properly.

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Data availability

No new observational data were generated for this work. All quantitative claims derive from the cited literature; analysis conventions for the proposed programs will be released with the respective pipeline papers. The pre-registration documents (hypotheses, trigger matrix, thresholds, and analysis plans, as fixed by Tables 6 and 5) are maintained at `https://omegacentauri.me`.

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